

# 1 SAT Lecture 2026 - Exercise 3.3 - Stardew Valley Soil Tilling Problem

470 instances  
Running 48 threads in parallel on EPYC 9684X 96-Core  
with 300s timeout

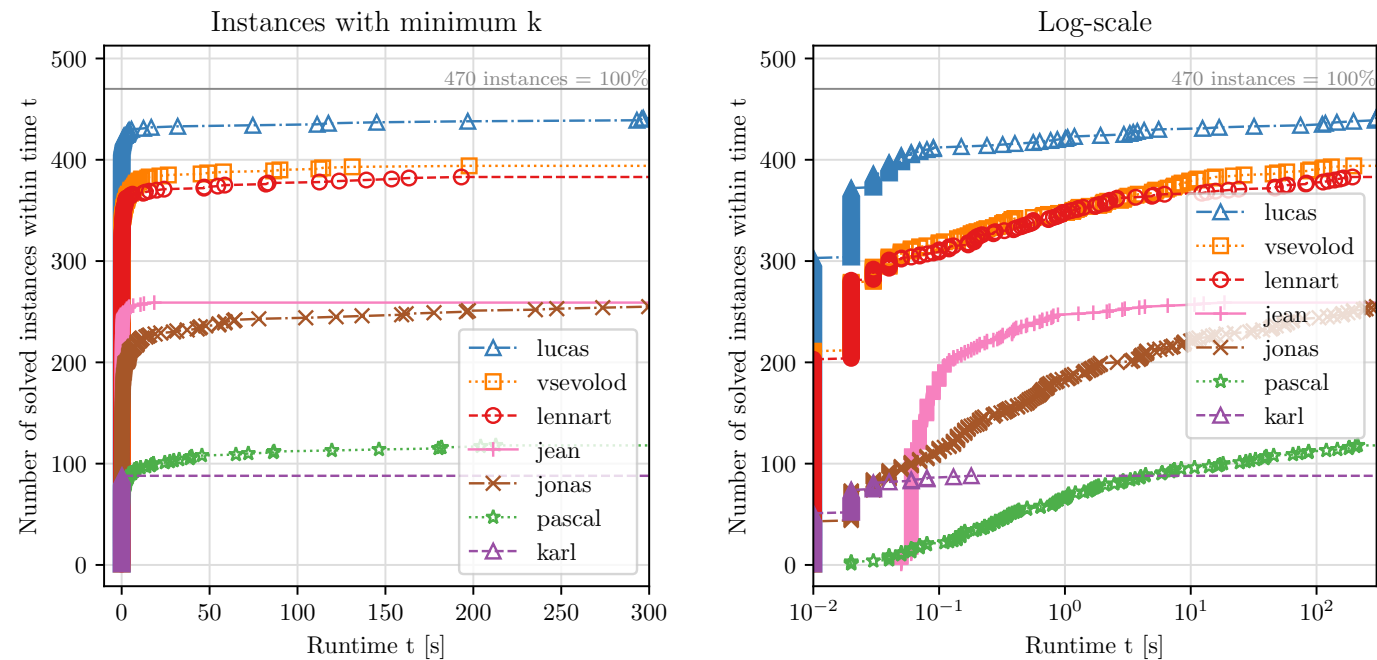


Figure 1

Table 1: Instances with at least 3 reported results: 434/470. For each such instance, compare k relative to others

| Student  | k minimal | k not minimal |
|----------|-----------|---------------|
| jean     | 256       | 178           |
| lennart  | 383       | 0             |
| pascal   | 118       | 0             |
| jonas    | 255       | 0             |
| vsevolod | 394       | 0             |
| lucas    | 433       | 0             |
| karl     | 77        | 341           |

Table 2: runtime and k per instance

| Instance                | jean       | jonas        | karl       | lennart    | lucas      | pascal      | vsevolod   |
|-------------------------|------------|--------------|------------|------------|------------|-------------|------------|
| gen-w10-h3-rep1.sdvstp  | 0.06<br>5  | 0.02<br>5    | 0.01<br>13 | 0.01<br>5  | 0.01<br>5  | 0.50<br>5   | 0.01<br>5  |
| gen-w10-h3-rep2.sdvstp  | 0.09<br>7  | 0.03<br>7    | 0.01<br>13 | 0.01<br>7  | 0.01<br>7  | 2.90<br>7   | 0.01<br>7  |
| gen-w10-h3-rep3.sdvstp  | 0.09<br>7  | 0.04<br>7    | 0.01<br>13 | 0.01<br>7  | 0.01<br>7  | 7.27<br>7   | 0.01<br>7  |
| gen-w10-h3-rep4.sdvstp  | 0.05<br>6  | 0.03<br>6    | 0.01<br>14 | 0.01<br>6  | 0.01<br>6  | 1.07<br>6   | 0.01<br>6  |
| gen-w10-h3-rep5.sdvstp  | 0.06<br>4  | 0.02<br>4    | 0.01<br>4  | –<br>–     | 0.01<br>4  | 0.36<br>4   | 0.01<br>4  |
| gen-w10-h4-rep1.sdvstp  | 0.07<br>8  | 0.15<br>8    | 0.01<br>17 | 0.01<br>8  | 0.01<br>8  | 180.22<br>8 | 0.01<br>8  |
| gen-w10-h4-rep2.sdvstp  | 0.07<br>8  | 0.14<br>8    | 0.01<br>17 | 0.01<br>8  | 0.01<br>8  | 45.65<br>8  | 0.01<br>8  |
| gen-w10-h4-rep3.sdvstp  | 0.06<br>8  | 0.14<br>8    | 0.01<br>16 | 0.01<br>8  | 0.01<br>8  | 72.02<br>8  | 0.01<br>8  |
| gen-w10-h4-rep4.sdvstp  | 0.06<br>5  | 0.02<br>5    | 0.01<br>10 | 0.01<br>5  | 0.01<br>5  | 1.42<br>5   | 0.01<br>5  |
| gen-w10-h4-rep5.sdvstp  | 0.06<br>8  | 0.12<br>8    | 0.01<br>15 | 0.01<br>8  | 0.01<br>8  | 86.63<br>8  | 0.01<br>8  |
| gen-w10-h5-rep1.sdvstp  | 0.06<br>8  | 0.18<br>8    | 0.01<br>25 | 0.01<br>8  | 0.02<br>8  | –<br>–      | 0.01<br>8  |
| gen-w10-h5-rep2.sdvstp  | 0.07<br>10 | 0.36<br>9    | 0.01<br>20 | 0.01<br>9  | 0.01<br>9  | –<br>–      | 0.01<br>9  |
| gen-w10-h5-rep3.sdvstp  | 0.11<br>9  | 0.17<br>9    | 0.01<br>17 | 0.01<br>9  | 0.01<br>9  | –<br>–      | 0.01<br>9  |
| gen-w10-h5-rep4.sdvstp  | 0.08<br>10 | 0.31<br>10   | 0.01<br>25 | 0.01<br>10 | 0.01<br>10 | –<br>–      | 0.01<br>10 |
| gen-w10-h5-rep5.sdvstp  | 0.06<br>9  | 0.27<br>8    | 0.01<br>29 | 0.02<br>8  | 0.01<br>8  | –<br>–      | 0.01<br>8  |
| gen-w10-h6-rep1.sdvstp  | 0.06<br>5  | 0.10<br>5    | 0.01<br>21 | 0.01<br>5  | 0.02<br>5  | 8.40<br>5   | 0.01<br>5  |
| gen-w10-h6-rep2.sdvstp  | 0.06<br>9  | 0.63<br>9    | 0.01<br>24 | 0.01<br>9  | 0.01<br>9  | –<br>–      | 0.01<br>9  |
| gen-w10-h6-rep3.sdvstp  | 0.06<br>9  | 0.23<br>9    | 0.01<br>24 | 0.01<br>9  | 0.01<br>9  | –<br>–      | 0.01<br>9  |
| gen-w10-h6-rep4.sdvstp  | 0.06<br>8  | 0.20<br>8    | 0.01<br>20 | 0.01<br>8  | 0.01<br>8  | –<br>–      | 0.01<br>8  |
| gen-w10-h6-rep5.sdvstp  | 0.06<br>8  | 0.48<br>8    | 0.01<br>25 | 0.01<br>8  | 0.01<br>8  | –<br>–      | 0.01<br>8  |
| gen-w10-h7-rep1.sdvstp  | 0.07<br>12 | 2.61<br>11   | 0.01<br>32 | 0.02<br>11 | 0.01<br>11 | –<br>–      | 0.02<br>11 |
| gen-w10-h7-rep2.sdvstp  | 0.10<br>12 | 1.13<br>11   | 0.01<br>30 | 0.01<br>11 | 0.01<br>11 | –<br>–      | 0.01<br>11 |
| gen-w10-h7-rep3.sdvstp  | 0.08<br>11 | 1.72<br>11   | 0.01<br>40 | 0.02<br>11 | 0.02<br>11 | –<br>–      | 0.02<br>11 |
| gen-w10-h7-rep4.sdvstp  | 0.10<br>13 | 15.87<br>13  | 0.01<br>34 | 0.01<br>13 | 0.01<br>13 | –<br>–      | 0.02<br>13 |
| gen-w10-h7-rep5.sdvstp  | 0.10<br>12 | 3.84<br>12   | 0.01<br>31 | 0.01<br>12 | 0.01<br>12 | –<br>–      | 0.01<br>12 |
| gen-w10-h9-rep1.sdvstp  | 0.15<br>12 | 4.11<br>10   | 0.01<br>50 | 0.02<br>10 | 0.03<br>10 | –<br>–      | 0.02<br>10 |
| gen-w10-h9-rep2.sdvstp  | 0.15<br>15 | –<br>–       | 0.01<br>61 | 0.02<br>15 | 0.01<br>15 | –<br>–      | 0.02<br>15 |
| gen-w10-h9-rep3.sdvstp  | 0.10<br>13 | 14.94<br>13  | 0.01<br>49 | 0.03<br>13 | 0.01<br>13 | –<br>–      | 0.04<br>13 |
| gen-w10-h9-rep4.sdvstp  | 0.11<br>15 | 247.45<br>14 | 0.01<br>39 | 0.02<br>14 | 0.01<br>14 | –<br>–      | 0.02<br>14 |
| gen-w10-h9-rep5.sdvstp  | 0.11<br>13 | 8.77<br>13   | 0.01<br>39 | 0.02<br>13 | 0.01<br>13 | –<br>–      | 0.02<br>13 |
| gen-w12-h11-rep1.sdvstp | 0.10<br>13 | 44.76<br>13  | 0.01<br>55 | 0.02<br>13 | 0.02<br>13 | –<br>–      | 0.02<br>13 |
| gen-w12-h11-rep2.sdvstp | 0.23<br>17 | –<br>–       | 0.01<br>71 | 0.03<br>15 | 0.01<br>15 | –<br>–      | 0.04<br>15 |
| gen-w12-h11-rep3.sdvstp | 0.18<br>16 | –<br>–       | 0.01<br>50 | 0.02<br>16 | 0.02<br>16 | –<br>–      | 0.02<br>16 |
| gen-w12-h11-rep4.sdvstp | 0.32<br>19 | –<br>–       | 0.01<br>66 | 0.02<br>19 | 0.01<br>19 | –<br>–      | 0.03<br>19 |
| gen-w12-h11-rep5.sdvstp | 0.38<br>19 | –<br>–       | 0.01<br>50 | 0.02<br>19 | 0.01<br>19 | –<br>–      | 0.02<br>19 |
| gen-w12-h3-rep1.sdvstp  | 0.06       | 0.10         | 0.01       | 0.01       | 0.01       | 17.08       | 0.01       |

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Table 2: runtime and k per instance (continued)

| Instance                | jean       | jonas        | karl        | lennart     | lucas      | pascal     | vsevolod     |
|-------------------------|------------|--------------|-------------|-------------|------------|------------|--------------|
|                         | 8          | 8            | 17          | 8           | 8          | 8          | 8            |
| gen-w12-h3-rep2.sdvstp  | 0.06<br>6  | 0.02<br>6    | 0.01<br>10  | 0.01<br>6   | 0.01<br>6  | 1.71<br>6  | 0.01<br>6    |
| gen-w12-h3-rep3.sdvstp  | 0.06<br>8  | 0.06<br>8    | 0.01<br>14  | 0.01<br>8   | 0.01<br>8  | 64.97<br>8 | 0.01<br>8    |
| gen-w12-h3-rep4.sdvstp  | 0.07<br>8  | 0.04<br>8    | 0.01<br>13  | 0.01<br>8   | 0.01<br>8  | 13.96<br>8 | 0.01<br>8    |
| gen-w12-h3-rep5.sdvstp  | 0.06<br>5  | 0.02<br>5    | 0.01<br>5   | 0.01<br>5   | 0.01<br>5  | 0.79<br>5  | 0.01<br>5    |
| gen-w12-h4-rep1.sdvstp  | 0.06<br>10 | 0.28<br>10   | 0.01<br>15  | 0.01<br>10  | 0.01<br>10 | —<br>—     | 0.01<br>10   |
| gen-w12-h4-rep2.sdvstp  | 0.08<br>11 | 0.60<br>11   | 0.01<br>27  | 0.02<br>11  | 0.01<br>11 | —<br>—     | 0.01<br>11   |
| gen-w12-h4-rep3.sdvstp  | 0.08<br>11 | 0.81<br>11   | 0.01<br>34  | 0.01<br>11  | 0.01<br>11 | —<br>—     | 0.02<br>11   |
| gen-w12-h4-rep4.sdvstp  | 0.07<br>8  | 0.15<br>8    | 0.01<br>17  | 0.01<br>8   | 0.01<br>8  | —<br>—     | 0.08<br>8    |
| gen-w12-h4-rep5.sdvstp  | 0.06<br>8  | 0.08<br>8    | 0.01<br>15  | 0.01<br>8   | 0.01<br>8  | —<br>—     | 0.01<br>8    |
| gen-w12-h5-rep1.sdvstp  | 0.07<br>9  | 0.38<br>9    | 0.01<br>23  | 0.01<br>9   | 0.01<br>9  | —<br>—     | 0.01<br>9    |
| gen-w12-h5-rep2.sdvstp  | 0.08<br>8  | 0.21<br>8    | 0.01<br>34  | 0.01<br>8   | 0.01<br>8  | —<br>—     | 0.01<br>8    |
| gen-w12-h5-rep3.sdvstp  | 0.08<br>14 | 235.29<br>14 | 0.01<br>27  | 0.01<br>14  | 0.01<br>14 | —<br>—     | 0.01<br>14   |
| gen-w12-h5-rep4.sdvstp  | 0.07<br>10 | 0.35<br>9    | 0.01<br>23  | 0.01<br>9   | 0.01<br>9  | —<br>—     | 0.01<br>9    |
| gen-w12-h5-rep5.sdvstp  | 0.06<br>6  | 0.13<br>6    | 0.01<br>26  | 0.01<br>6   | 0.01<br>6  | 27.42<br>6 | 0.01<br>6    |
| gen-w12-h6-rep1.sdvstp  | 0.09<br>10 | 0.50<br>10   | 0.01<br>24  | 0.01<br>10  | 0.01<br>10 | —<br>—     | 0.01<br>10   |
| gen-w12-h6-rep2.sdvstp  | 0.23<br>18 | —<br>—       | 0.01<br>34  | 0.01<br>17  | 0.01<br>17 | —<br>—     | 0.01<br>17   |
| gen-w12-h6-rep3.sdvstp  | 0.08<br>13 | 7.24<br>13   | 0.01<br>35  | 0.02<br>13  | 0.01<br>13 | —<br>—     | 0.01<br>13   |
| gen-w12-h6-rep4.sdvstp  | 0.07<br>11 | 1.05<br>11   | 0.01<br>37  | 0.01<br>11  | 0.01<br>11 | —<br>—     | 0.02<br>11   |
| gen-w12-h6-rep5.sdvstp  | 0.08<br>10 | 0.56<br>10   | 0.01<br>35  | 0.02<br>10  | 0.01<br>10 | —<br>—     | 0.02<br>10   |
| gen-w12-h7-rep1.sdvstp  | 0.17<br>16 | —<br>—       | 0.01<br>34  | 0.02<br>16  | 0.01<br>16 | —<br>—     | 0.01<br>16   |
| gen-w12-h7-rep2.sdvstp  | 0.10<br>13 | 4.47<br>12   | 0.01<br>58  | 0.03<br>12  | 0.05<br>12 | —<br>—     | 0.07<br>12   |
| gen-w12-h7-rep3.sdvstp  | 0.10<br>13 | 42.97<br>13  | 0.01<br>34  | 0.02<br>13  | 0.02<br>13 | —<br>—     | 0.02<br>13   |
| gen-w12-h7-rep4.sdvstp  | 0.12<br>15 | 177.63<br>15 | 0.01<br>41  | 0.02<br>15  | 0.02<br>15 | —<br>—     | 0.02<br>15   |
| gen-w12-h7-rep5.sdvstp  | 0.12<br>14 | —<br>—       | 0.01<br>47  | 0.02<br>14  | 0.01<br>14 | —<br>—     | 0.02<br>14   |
| gen-w12-h9-rep1.sdvstp  | 0.29<br>18 | —<br>—       | 0.01<br>44  | 0.02<br>18  | 0.01<br>18 | —<br>—     | 0.03<br>18   |
| gen-w12-h9-rep2.sdvstp  | 0.14<br>16 | 159.23<br>15 | 0.01<br>77  | 0.20<br>15  | 0.06<br>15 | —<br>—     | 0.22<br>15   |
| gen-w12-h9-rep3.sdvstp  | 0.16<br>16 | —<br>—       | 0.01<br>53  | 0.02<br>15  | 0.01<br>15 | —<br>—     | 0.03<br>15   |
| gen-w12-h9-rep4.sdvstp  | 0.38<br>19 | —<br>—       | 0.01<br>48  | 0.02<br>18  | 0.01<br>18 | —<br>—     | 0.02<br>18   |
| gen-w12-h9-rep5.sdvstp  | 0.14<br>15 | 55.39<br>14  | 0.01<br>43  | 0.02<br>14  | 0.01<br>14 | —<br>—     | 0.02<br>14   |
| gen-w14-h11-rep1.sdvstp | 0.38<br>18 | —<br>—       | 0.01<br>72  | 0.06<br>18  | 0.01<br>18 | —<br>—     | 0.04<br>18   |
| gen-w14-h11-rep2.sdvstp | 0.24<br>18 | —<br>—       | 0.01<br>75  | 0.03<br>16  | 0.01<br>16 | —<br>—     | 0.04<br>16   |
| gen-w14-h11-rep3.sdvstp | 0.82<br>24 | —<br>—       | —<br>—      | 0.97<br>22  | 0.01<br>22 | —<br>—     | 0.32<br>22   |
| gen-w14-h11-rep4.sdvstp | 1.28<br>25 | —<br>—       | 0.01<br>77  | 0.04<br>25  | 0.03<br>25 | —<br>—     | 0.29<br>25   |
| gen-w14-h11-rep5.sdvstp | 1.80<br>31 | —<br>—       | 0.01<br>110 | 83.24<br>28 | 0.02<br>28 | —<br>—     | 112.18<br>28 |
| gen-w14-h14-rep1.sdvstp | 1.98<br>32 | —<br>—       | 0.01<br>134 | —<br>—      | 1.05<br>29 | —<br>—     | —<br>—       |

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Table 2: runtime and k per instance (continued)

| Instance                | jean       | jonas        | karl        | lennart    | lucas      | pascal      | vsevolod   |
|-------------------------|------------|--------------|-------------|------------|------------|-------------|------------|
| gen-w14-h14-rep2.sdvstp | 1.02<br>25 | —<br>—       | 0.01<br>92  | 0.19<br>24 | 0.03<br>24 | —<br>—      | 0.37<br>24 |
| gen-w14-h14-rep3.sdvstp | 0.38<br>19 | —<br>—       | 0.01<br>100 | 0.07<br>19 | 0.04<br>19 | —<br>—      | 0.27<br>19 |
| gen-w14-h14-rep4.sdvstp | 2.56<br>34 | —<br>—       | 0.01<br>95  | 0.39<br>32 | 0.01<br>32 | —<br>—      | 3.83<br>32 |
| gen-w14-h14-rep5.sdvstp | 1.69<br>30 | —<br>—       | 0.01<br>98  | 1.77<br>28 | 0.01<br>28 | —<br>—      | 9.83<br>28 |
| gen-w14-h3-rep1.sdvstp  | 0.06<br>9  | 0.13<br>8    | 0.01<br>26  | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.01<br>8  |
| gen-w14-h3-rep2.sdvstp  | 0.07<br>8  | 0.12<br>8    | 0.01<br>23  | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.01<br>8  |
| gen-w14-h3-rep3.sdvstp  | 0.06<br>5  | 0.04<br>5    | 0.01<br>19  | 0.01<br>5  | 0.02<br>5  | 2.47<br>5   | 0.01<br>5  |
| gen-w14-h3-rep4.sdvstp  | 0.06<br>11 | 0.52<br>11   | 0.01<br>14  | 0.01<br>11 | 0.01<br>11 | —<br>—      | 0.06<br>11 |
| gen-w14-h3-rep5.sdvstp  | 0.06<br>8  | 0.11<br>8    | 0.01<br>19  | 0.01<br>8  | 0.01<br>8  | 119.54<br>8 | 0.01<br>8  |
| gen-w14-h4-rep1.sdvstp  | 0.09<br>7  | 0.09<br>7    | 0.01<br>11  | 0.01<br>7  | 0.01<br>7  | 39.80<br>7  | 0.01<br>7  |
| gen-w14-h4-rep2.sdvstp  | 0.06<br>9  | 0.27<br>9    | 0.01<br>34  | 0.02<br>9  | 0.01<br>9  | —<br>—      | 0.01<br>9  |
| gen-w14-h4-rep3.sdvstp  | 0.06<br>9  | 0.21<br>8    | 0.01<br>32  | 0.02<br>8  | 0.02<br>8  | —<br>—      | 0.01<br>8  |
| gen-w14-h4-rep4.sdvstp  | 0.07<br>9  | 0.14<br>8    | 0.01<br>20  | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.01<br>8  |
| gen-w14-h4-rep5.sdvstp  | 0.11<br>13 | 6.42<br>13   | 0.01<br>21  | 0.01<br>13 | 0.01<br>13 | —<br>—      | 0.01<br>13 |
| gen-w14-h5-rep1.sdvstp  | 0.07<br>10 | 0.66<br>10   | 0.01<br>36  | 0.01<br>10 | 0.01<br>10 | —<br>—      | 0.01<br>10 |
| gen-w14-h5-rep2.sdvstp  | 0.08<br>12 | 4.30<br>12   | 0.01<br>30  | 0.01<br>12 | 0.01<br>12 | —<br>—      | 0.01<br>12 |
| gen-w14-h5-rep3.sdvstp  | 0.07<br>11 | 0.54<br>9    | 0.01<br>30  | 0.01<br>9  | 0.01<br>9  | —<br>—      | 0.01<br>9  |
| gen-w14-h5-rep4.sdvstp  | 0.12<br>15 | —<br>—       | 0.01<br>35  | 0.01<br>15 | 0.01<br>15 | —<br>—      | 0.01<br>15 |
| gen-w14-h5-rep5.sdvstp  | 0.07<br>10 | 0.58<br>9    | 0.01<br>39  | 0.02<br>9  | 0.02<br>9  | —<br>—      | 0.02<br>9  |
| gen-w14-h6-rep1.sdvstp  | 0.09<br>12 | 4.46<br>12   | 0.01<br>37  | 0.01<br>12 | 0.01<br>12 | —<br>—      | 0.02<br>12 |
| gen-w14-h6-rep2.sdvstp  | 0.12<br>14 | 104.69<br>14 | 0.01<br>42  | 0.02<br>14 | 0.01<br>14 | —<br>—      | 0.01<br>14 |
| gen-w14-h6-rep3.sdvstp  | 0.10<br>15 | 31.37<br>14  | 0.01<br>40  | 0.01<br>14 | 0.01<br>14 | —<br>—      | 0.02<br>14 |
| gen-w14-h6-rep4.sdvstp  | 0.08<br>11 | 1.60<br>10   | 0.01<br>42  | 0.02<br>10 | 0.01<br>10 | —<br>—      | 0.02<br>10 |
| gen-w14-h6-rep5.sdvstp  | 0.08<br>11 | 5.15<br>11   | 0.01<br>35  | 0.01<br>11 | 0.01<br>11 | —<br>—      | 0.01<br>11 |
| gen-w14-h7-rep1.sdvstp  | 0.13<br>15 | 58.84<br>14  | 0.01<br>46  | 0.02<br>14 | 0.01<br>14 | —<br>—      | 0.02<br>14 |
| gen-w14-h7-rep2.sdvstp  | 0.27<br>19 | —<br>—       | 0.01<br>45  | 0.02<br>17 | 0.01<br>17 | —<br>—      | 0.02<br>17 |
| gen-w14-h7-rep3.sdvstp  | 0.19<br>19 | —<br>—       | 0.01<br>43  | 0.02<br>19 | 0.01<br>19 | —<br>—      | 0.02<br>19 |
| gen-w14-h7-rep4.sdvstp  | 0.23<br>18 | —<br>—       | 0.01<br>47  | 0.03<br>17 | 0.01<br>17 | —<br>—      | 0.02<br>17 |
| gen-w14-h7-rep5.sdvstp  | 0.16<br>16 | 273.60<br>15 | 0.01<br>47  | 0.02<br>15 | 0.01<br>15 | —<br>—      | 0.02<br>15 |
| gen-w14-h9-rep1.sdvstp  | 0.17<br>19 | —<br>—       | 0.01<br>57  | 0.02<br>18 | 0.01<br>18 | —<br>—      | 0.02<br>18 |
| gen-w14-h9-rep2.sdvstp  | 0.39<br>19 | —<br>—       | 0.01<br>85  | 0.53<br>19 | 0.02<br>19 | —<br>—      | 1.03<br>19 |
| gen-w14-h9-rep3.sdvstp  | 0.40<br>18 | —<br>—       | 0.01<br>62  | 0.10<br>18 | 0.01<br>18 | —<br>—      | 0.10<br>18 |
| gen-w14-h9-rep4.sdvstp  | 0.41<br>19 | —<br>—       | 0.01<br>77  | 0.02<br>19 | 0.01<br>19 | —<br>—      | 0.06<br>19 |
| gen-w14-h9-rep5.sdvstp  | 0.32<br>16 | —<br>—       | 0.01<br>79  | 0.28<br>16 | 1.19<br>16 | —<br>—      | 0.33<br>16 |
| gen-w16-h11-rep1.sdvstp | 1.37<br>28 | —<br>—       | 0.01<br>77  | 0.02<br>26 | 0.01<br>26 | —<br>—      | 0.18<br>26 |
| gen-w16-h11-rep2.sdvstp | 0.65       | —            | 0.01        | 0.45       | 0.02       | —           | 1.15       |

continued on next page

Table 2: runtime and k per instance (continued)

| Instance                | jean | jonas  | karl | lennart | lucas | pascal | vsevolod |
|-------------------------|------|--------|------|---------|-------|--------|----------|
|                         | 21   | —      | 94   | 19      | 19    | —      | 19       |
| gen-w16-h11-rep3.sdvstp | 0.89 | —      | —    | 12.24   | 4.26  | —      | 2.94     |
|                         | 24   | —      | —    | 21      | 21    | —      | 21       |
| gen-w16-h11-rep4.sdvstp | 1.37 | —      | 0.01 | 0.33    | 0.03  | —      | 7.42     |
|                         | 27   | —      | 95   | 25      | 25    | —      | 25       |
| gen-w16-h11-rep5.sdvstp | 1.16 | —      | 0.01 | 0.12    | 0.06  | —      | 0.67     |
|                         | 25   | —      | 89   | 24      | 24    | —      | 24       |
| gen-w16-h14-rep1.sdvstp | 4.32 | —      | 0.01 | 0.56    | 0.02  | —      | 4.96     |
|                         | 38   | —      | 38   | 38      | 38    | —      | 38       |
| gen-w16-h14-rep2.sdvstp | 4.17 | —      | 0.01 | 0.89    | 0.04  | —      | 0.38     |
|                         | 39   | —      | 38   | 38      | 38    | —      | 38       |
| gen-w16-h14-rep3.sdvstp | 2.26 | —      | 0.01 | 2.59    | 0.08  | —      | 6.61     |
|                         | 31   | —      | 31   | 31      | 31    | —      | 31       |
| gen-w16-h14-rep4.sdvstp | 6.31 | —      | 0.02 | 82.03   | 0.04  | —      | 131.12   |
|                         | 38   | —      | 37   | 36      | 36    | —      | 36       |
| gen-w16-h14-rep5.sdvstp | 3.69 | —      | 0.02 | 19.93   | 0.02  | —      | 8.54     |
|                         | 31   | —      | 29   | 29      | 29    | —      | 29       |
| gen-w16-h3-rep1.sdvstp  | 0.08 | 1.54   | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                         | 12   | 11     | 23   | 11      | 11    | —      | 11       |
| gen-w16-h3-rep2.sdvstp  | 0.07 | 0.67   | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                         | 11   | 11     | 19   | 11      | 11    | —      | 11       |
| gen-w16-h3-rep3.sdvstp  | 0.06 | 0.02   | 0.01 | 0.01    | 0.01  | 1.85   | 0.01     |
|                         | 4    | 4      | 13   | 4       | 4     | 4      | 4        |
| gen-w16-h3-rep4.sdvstp  | 0.06 | 0.04   | 0.01 | 0.01    | 0.01  | 5.93   | 0.01     |
|                         | 7    | 6      | 20   | 6       | 6     | 6      | 6        |
| gen-w16-h3-rep5.sdvstp  | 0.08 | 0.52   | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                         | 11   | 10     | 29   | 10      | 10    | —      | 10       |
| gen-w16-h4-rep1.sdvstp  | 0.06 | 0.42   | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                         | 8    | 8      | 27   | 8       | 8     | —      | 8        |
| gen-w16-h4-rep2.sdvstp  | 0.10 | 3.35   | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                         | 12   | 12     | 31   | 12      | 12    | —      | 12       |
| gen-w16-h4-rep3.sdvstp  | 0.07 | 4.03   | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                         | 12   | 12     | 25   | 12      | 12    | —      | 12       |
| gen-w16-h4-rep4.sdvstp  | 0.08 | 1.43   | 0.01 | 0.01    | 0.01  | —      | 0.02     |
|                         | 11   | 11     | 37   | 11      | 11    | —      | 11       |
| gen-w16-h4-rep5.sdvstp  | 0.09 | 55.16  | 0.01 | 0.01    | 0.02  | —      | 0.01     |
|                         | 15   | 14     | 26   | 14      | 14    | —      | 14       |
| gen-w16-h5-rep1.sdvstp  | 0.23 | —      | 0.01 | 0.02    | 0.01  | —      | 0.02     |
|                         | 17   | —      | 37   | 17      | 17    | —      | 17       |
| gen-w16-h5-rep2.sdvstp  | 0.08 | 0.62   | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                         | 10   | 10     | 38   | 10      | 10    | —      | 10       |
| gen-w16-h5-rep3.sdvstp  | 0.07 | 0.74   | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                         | 11   | 11     | 33   | 11      | 11    | —      | 11       |
| gen-w16-h5-rep4.sdvstp  | 0.27 | —      | 0.01 | 0.02    | 0.01  | —      | 0.02     |
|                         | 18   | —      | 46   | 17      | 17    | —      | 17       |
| gen-w16-h5-rep5.sdvstp  | 0.10 | 1.91   | 0.01 | 0.02    | 0.01  | —      | 0.02     |
|                         | 10   | 10     | 36   | 10      | 10    | —      | 10       |
| gen-w16-h6-rep1.sdvstp  | 0.10 | 4.02   | 0.01 | 0.02    | 0.01  | —      | 0.02     |
|                         | 12   | 12     | 39   | 12      | 12    | —      | 12       |
| gen-w16-h6-rep2.sdvstp  | 0.13 | —      | 0.01 | 0.02    | 0.01  | —      | 0.02     |
|                         | 16   | —      | 45   | 15      | 15    | —      | 15       |
| gen-w16-h6-rep3.sdvstp  | 0.28 | —      | 0.01 | 0.04    | 0.01  | —      | 0.05     |
|                         | 21   | —      | 52   | 20      | 20    | —      | 20       |
| gen-w16-h6-rep4.sdvstp  | 0.17 | 136.70 | 0.01 | 0.01    | 0.02  | —      | 0.02     |
|                         | 16   | 15     | 41   | 15      | 15    | —      | 15       |
| gen-w16-h6-rep5.sdvstp  | 0.11 | 64.44  | 0.01 | 0.02    | 0.01  | —      | 0.03     |
|                         | 14   | 14     | 47   | 14      | 14    | —      | 14       |
| gen-w16-h7-rep1.sdvstp  | 0.18 | —      | 0.01 | 0.02    | 0.01  | —      | 0.03     |
|                         | 17   | —      | 48   | 17      | 17    | —      | 17       |
| gen-w16-h7-rep2.sdvstp  | 0.13 | 10.47  | 0.01 | 0.05    | 0.04  | —      | 0.06     |
|                         | 13   | 12     | 64   | 12      | 12    | —      | 12       |
| gen-w16-h7-rep3.sdvstp  | 0.13 | 6.98   | 0.01 | 0.04    | 0.01  | —      | 0.03     |
|                         | 14   | 12     | 53   | 12      | 12    | —      | 12       |
| gen-w16-h7-rep4.sdvstp  | 0.63 | —      | 0.01 | 0.02    | 0.01  | —      | 0.03     |
|                         | 23   | —      | 57   | 23      | 23    | —      | 23       |
| gen-w16-h7-rep5.sdvstp  | 0.08 | 37.26  | 0.01 | 0.02    | 0.01  | —      | 0.02     |
|                         | 13   | 13     | 43   | 13      | 13    | —      | 13       |
| gen-w16-h9-rep1.sdvstp  | 0.36 | —      | 0.01 | 0.04    | 0.01  | —      | 0.05     |
|                         | 18   | —      | 62   | 17      | 17    | —      | 17       |
| gen-w16-h9-rep2.sdvstp  | 0.41 | —      | 0.01 | 0.42    | 1.01  | —      | 0.24     |
|                         | 19   | —      | 78   | 18      | 18    | —      | 18       |

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Table 2: runtime and k per instance (continued)

| Instance                | jean        | jonas        | karl       | lennart     | lucas        | pascal    | vsevolod    |
|-------------------------|-------------|--------------|------------|-------------|--------------|-----------|-------------|
| gen-w16-h9-rep3.sdvstp  | 0.66<br>20  | —<br>—       | 0.01<br>88 | 2.10<br>20  | 0.02<br>20   | —<br>—    | 0.18<br>20  |
| gen-w16-h9-rep4.sdvstp  | 0.27<br>18  | —<br>—       | 0.01<br>87 | 0.19<br>16  | 0.04<br>16   | —<br>—    | 0.16<br>16  |
| gen-w16-h9-rep5.sdvstp  | 0.19<br>16  | —<br>—       | 0.01<br>60 | 0.02<br>15  | 0.01<br>15   | —<br>—    | 0.03<br>15  |
| gen-w19-h11-rep1.sdvstp | 1.30<br>26  | —<br>—       | 0.01<br>25 | 15.75<br>25 | 1.92<br>25   | —<br>—    | 15.69<br>25 |
| gen-w19-h11-rep2.sdvstp | 3.05<br>30  | —<br>—       | —<br>—     | 0.27<br>28  | 0.02<br>28   | —<br>—    | 2.19<br>28  |
| gen-w19-h11-rep3.sdvstp | 1.57<br>31  | —<br>—       | 0.01<br>28 | 24.10<br>28 | 3.76<br>28   | —<br>—    | 11.09<br>28 |
| gen-w19-h11-rep4.sdvstp | 1.94<br>30  | —<br>—       | 0.02<br>29 | 58.62<br>29 | 0.24<br>29   | —<br>—    | 86.45<br>29 |
| gen-w19-h11-rep5.sdvstp | 2.43<br>31  | —<br>—       | 0.01<br>29 | 1.74<br>29  | 0.02<br>29   | —<br>—    | 5.21<br>29  |
| gen-w19-h14-rep1.sdvstp | 7.15<br>39  | —<br>—       | 0.02<br>39 | —<br>—      | 295.67<br>38 | —<br>—    | —<br>—      |
| gen-w19-h14-rep2.sdvstp | 12.55<br>45 | —<br>—       | 0.02<br>42 | —<br>—      | 5.56<br>42   | —<br>—    | —<br>—      |
| gen-w19-h14-rep3.sdvstp | 2.85<br>35  | —<br>—       | 0.02<br>32 | —<br>—      | 0.04<br>32   | —<br>—    | —<br>—      |
| gen-w19-h14-rep4.sdvstp | 4.66<br>40  | —<br>—       | 0.04<br>36 | —<br>—      | 0.05<br>35   | —<br>—    | —<br>—      |
| gen-w19-h14-rep5.sdvstp | 4.13<br>37  | —<br>—       | 0.01<br>37 | —<br>—      | 0.04<br>36   | —<br>—    | —<br>—      |
| gen-w19-h18-rep1.sdvstp | 18.54<br>52 | —<br>—       | 0.04<br>52 | —<br>—      | —<br>—       | —<br>—    | —<br>—      |
| gen-w19-h18-rep2.sdvstp | 6.56<br>41  | —<br>—       | 0.03<br>42 | —<br>—      | —<br>—       | —<br>—    | —<br>—      |
| gen-w19-h18-rep3.sdvstp | 23.23<br>56 | —<br>—       | 0.02<br>54 | —<br>—      | 0.77<br>53   | —<br>—    | —<br>—      |
| gen-w19-h18-rep4.sdvstp | 20.33<br>55 | —<br>—       | 0.02<br>53 | —<br>—      | 0.02<br>52   | —<br>—    | —<br>—      |
| gen-w19-h18-rep5.sdvstp | 17.15<br>47 | —<br>—       | 0.03<br>46 | —<br>—      | 0.02<br>45   | —<br>—    | —<br>—      |
| gen-w19-h3-rep1.sdvstp  | 0.09<br>11  | 4.45<br>11   | 0.01<br>32 | 0.02<br>11  | 0.01<br>11   | —<br>—    | 0.02<br>11  |
| gen-w19-h3-rep2.sdvstp  | 0.07<br>11  | 0.50<br>10   | 0.01<br>29 | 0.01<br>10  | 0.02<br>10   | —<br>—    | 0.01<br>10  |
| gen-w19-h3-rep3.sdvstp  | 0.06<br>5   | 0.04<br>5    | 0.01<br>23 | 0.01<br>5   | 0.01<br>5    | 4.01<br>5 | 0.01<br>5   |
| gen-w19-h3-rep4.sdvstp  | 0.08<br>12  | 0.72<br>11   | 0.01<br>24 | 0.01<br>11  | 0.01<br>11   | —<br>—    | 0.01<br>11  |
| gen-w19-h3-rep5.sdvstp  | 0.07<br>8   | 0.22<br>8    | 0.01<br>23 | 0.01<br>8   | 0.01<br>8    | —<br>—    | 0.01<br>8   |
| gen-w19-h4-rep1.sdvstp  | 0.09<br>10  | 0.64<br>10   | 0.01<br>28 | 0.01<br>10  | 0.01<br>10   | —<br>—    | 0.01<br>10  |
| gen-w19-h4-rep2.sdvstp  | 0.08<br>11  | 1.19<br>11   | 0.01<br>28 | 0.01<br>11  | 0.01<br>11   | —<br>—    | 0.01<br>11  |
| gen-w19-h4-rep3.sdvstp  | 0.09<br>13  | 13.80<br>13  | 0.01<br>27 | 0.01<br>13  | 0.01<br>13   | —<br>—    | 0.01<br>13  |
| gen-w19-h4-rep4.sdvstp  | 0.08<br>12  | 3.46<br>12   | 0.01<br>26 | 0.01<br>12  | 0.01<br>12   | —<br>—    | 0.01<br>12  |
| gen-w19-h4-rep5.sdvstp  | 0.08<br>12  | 0.87<br>11   | 0.01<br>22 | 0.01<br>11  | 0.01<br>11   | —<br>—    | 0.01<br>11  |
| gen-w19-h5-rep1.sdvstp  | 0.07<br>10  | 0.72<br>10   | 0.01<br>45 | 0.02<br>10  | 0.02<br>10   | —<br>—    | 0.02<br>10  |
| gen-w19-h5-rep2.sdvstp  | 0.13<br>15  | 196.02<br>15 | 0.01<br>37 | 0.02<br>15  | 0.01<br>15   | —<br>—    | 0.01<br>15  |
| gen-w19-h5-rep3.sdvstp  | 0.21<br>17  | —<br>—       | 0.01<br>50 | 0.02<br>17  | 0.01<br>17   | —<br>—    | 0.02<br>17  |
| gen-w19-h5-rep4.sdvstp  | 0.13<br>15  | 197.15<br>15 | 0.01<br>48 | 0.02<br>15  | 0.01<br>15   | —<br>—    | 0.02<br>15  |
| gen-w19-h5-rep5.sdvstp  | 0.20<br>18  | —<br>—       | 0.01<br>36 | 0.01<br>17  | 0.01<br>17   | —<br>—    | 0.01<br>17  |
| gen-w19-h6-rep1.sdvstp  | 0.09<br>13  | 10.99<br>13  | 0.01<br>42 | 0.01<br>13  | 0.01<br>13   | —<br>—    | 0.02<br>13  |
| gen-w19-h6-rep2.sdvstp  | 0.24<br>19  | —<br>—       | 0.01<br>49 | 0.03<br>19  | 0.01<br>19   | —<br>—    | 0.03<br>19  |
| gen-w19-h6-rep3.sdvstp  | 0.23        | —            | 0.01       | 0.03        | 0.02         | —         | 0.02        |

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Table 2: runtime and k per instance (continued)

| Instance                | jean  | jonas | karl | lennart | lucas  | pascal | vsevolod |
|-------------------------|-------|-------|------|---------|--------|--------|----------|
|                         | 17    | —     | 54   | 16      | 16     | —      | 16       |
| gen-w19-h6-rep4.sdvstp  | 0.49  | —     | 0.01 | 0.02    | 0.02   | —      | 0.02     |
|                         | 21    | —     | 50   | 21      | 21     | —      | 21       |
| gen-w19-h6-rep5.sdvstp  | 0.10  | 78.00 | 0.01 | 0.01    | 0.02   | —      | 0.01     |
|                         | 14    | 14    | 39   | 14      | 14     | —      | 14       |
| gen-w19-h7-rep1.sdvstp  | 0.66  | —     | —    | 2.16    | 3.28   | —      | 4.95     |
|                         | 21    | —     | —    | 21      | 21     | —      | 21       |
| gen-w19-h7-rep2.sdvstp  | 0.37  | —     | 0.01 | 0.17    | 0.03   | —      | 0.14     |
|                         | 19    | —     | 68   | 18      | 18     | —      | 18       |
| gen-w19-h7-rep3.sdvstp  | 0.48  | —     | 0.01 | 0.02    | 0.02   | —      | 0.02     |
|                         | 19    | —     | 52   | 18      | 18     | —      | 18       |
| gen-w19-h7-rep4.sdvstp  | 0.32  | —     | 0.01 | 0.13    | 0.02   | —      | 0.07     |
|                         | 18    | —     | 71   | 15      | 15     | —      | 15       |
| gen-w19-h7-rep5.sdvstp  | 0.61  | —     | —    | 0.10    | 0.01   | —      | 0.17     |
|                         | 21    | —     | —    | 19      | 19     | —      | 19       |
| gen-w19-h9-rep1.sdvstp  | 1.01  | —     | 0.01 | 1.06    | 0.01   | —      | 14.86    |
|                         | 26    | —     | 89   | 25      | 25     | —      | 25       |
| gen-w19-h9-rep2.sdvstp  | 0.81  | —     | 0.01 | 4.91    | 0.03   | —      | 2.91     |
|                         | 26    | —     | 97   | 26      | 26     | —      | 26       |
| gen-w19-h9-rep3.sdvstp  | 1.29  | —     | 0.01 | 2.18    | 0.01   | —      | 3.10     |
|                         | 29    | —     | 98   | 27      | 27     | —      | 27       |
| gen-w19-h9-rep4.sdvstp  | 0.80  | —     | —    | 0.22    | 0.02   | —      | 1.16     |
|                         | 21    | —     | —    | 21      | 21     | —      | 21       |
| gen-w19-h9-rep5.sdvstp  | 1.32  | —     | —    | 0.17    | 0.02   | —      | 1.30     |
|                         | 28    | —     | —    | 27      | 27     | —      | 27       |
| gen-w22-h11-rep1.sdvstp | 3.90  | —     | 0.02 | 139.17  | 0.02   | —      | 197.61   |
|                         | 33    | —     | 33   | 33      | 33     | —      | 33       |
| gen-w22-h11-rep2.sdvstp | 2.06  | —     | 0.01 | 1.40    | 0.02   | —      | 1.86     |
|                         | 30    | —     | 29   | 29      | 29     | —      | 29       |
| gen-w22-h11-rep3.sdvstp | 4.21  | —     | 0.02 | 163.39  | 0.04   | —      | —        |
|                         | 33    | —     | 32   | 32      | 32     | —      | —        |
| gen-w22-h11-rep4.sdvstp | 3.15  | —     | 0.02 | 193.18  | 0.02   | —      | —        |
|                         | 35    | —     | 31   | 31      | 31     | —      | —        |
| gen-w22-h11-rep5.sdvstp | 4.30  | —     | 0.02 | —       | 0.32   | —      | —        |
|                         | 41    | —     | 40   | —       | 40     | —      | —        |
| gen-w22-h14-rep1.sdvstp | 15.23 | —     | 0.03 | —       | 0.56   | —      | —        |
|                         | 49    | —     | 47   | —       | 47     | —      | —        |
| gen-w22-h14-rep2.sdvstp | 13.16 | —     | —    | —       | —      | —      | —        |
|                         | 44    | —     | —    | —       | —      | —      | —        |
| gen-w22-h14-rep3.sdvstp | 3.99  | —     | 0.02 | —       | 0.02   | —      | —        |
|                         | 40    | —     | 39   | —       | 39     | —      | —        |
| gen-w22-h14-rep4.sdvstp | 6.01  | —     | 0.02 | —       | 0.02   | —      | —        |
|                         | 43    | —     | 41   | —       | 41     | —      | —        |
| gen-w22-h14-rep5.sdvstp | 10.71 | —     | 0.03 | —       | 145.04 | —      | —        |
|                         | 46    | —     | 46   | —       | 46     | —      | —        |
| gen-w22-h18-rep1.sdvstp | 24.67 | —     | 0.08 | —       | —      | —      | —        |
|                         | 48    | —     | 46   | —       | —      | —      | —        |
| gen-w22-h18-rep2.sdvstp | 24.78 | —     | 0.05 | —       | 0.05   | —      | —        |
|                         | 51    | —     | 50   | —       | 49     | —      | —        |
| gen-w22-h18-rep3.sdvstp | 42.63 | —     | 0.03 | —       | —      | —      | —        |
|                         | 54    | —     | 53   | —       | —      | —      | —        |
| gen-w22-h18-rep4.sdvstp | 22.89 | —     | 0.03 | —       | 2.94   | —      | —        |
|                         | 51    | —     | 51   | —       | 47     | —      | —        |
| gen-w22-h18-rep5.sdvstp | 36.43 | —     | 0.08 | —       | —      | —      | —        |
|                         | 53    | —     | 51   | —       | —      | —      | —        |
| gen-w22-h3-rep1.sdvstp  | 0.06  | 0.17  | 0.01 | 0.01    | 0.01   | —      | 0.01     |
|                         | 9     | 9     | 20   | 9       | 9      | —      | 9        |
| gen-w22-h3-rep2.sdvstp  | 0.10  | 8.42  | 0.01 | 0.02    | 0.01   | —      | 0.01     |
|                         | 13    | 13    | 37   | 13      | 13     | —      | 13       |
| gen-w22-h3-rep3.sdvstp  | 0.06  | 0.07  | 0.01 | 0.01    | 0.01   | 16.49  | 0.01     |
|                         | 6     | 6     | 18   | 6       | 6      | 6      | 6        |
| gen-w22-h3-rep4.sdvstp  | 0.09  | 9.99  | 0.01 | 0.01    | 0.01   | —      | 0.01     |
|                         | 13    | 13    | 13   | 13      | 13     | —      | 13       |
| gen-w22-h3-rep5.sdvstp  | 0.12  | 3.97  | 0.01 | 0.01    | 0.01   | —      | 0.01     |
|                         | 12    | 12    | 30   | 12      | 12     | —      | 12       |
| gen-w22-h4-rep1.sdvstp  | 0.20  | —     | 0.01 | 0.03    | 0.01   | —      | 0.04     |
|                         | 17    | —     | 49   | 17      | 17     | —      | 17       |
| gen-w22-h4-rep2.sdvstp  | 0.10  | 5.68  | 0.01 | 0.01    | 0.01   | —      | 0.01     |
|                         | 12    | 12    | 32   | 12      | 12     | —      | 12       |
| gen-w22-h4-rep3.sdvstp  | 0.16  | —     | 0.01 | 0.02    | 0.01   | —      | 0.02     |
|                         | 16    | —     | 42   | 16      | 16     | —      | 16       |

continued on next page

Table 2: runtime and k per instance (continued)

| Instance                | jean        | jonas        | karl       | lennart      | lucas       | pascal | vsevolod    |
|-------------------------|-------------|--------------|------------|--------------|-------------|--------|-------------|
| gen-w22-h4-rep4.sdvstp  | 0.08<br>13  | 22.23<br>13  | 0.01<br>37 | 0.01<br>13   | 0.01<br>13  | —<br>— | 0.02<br>13  |
| gen-w22-h4-rep5.sdvstp  | 0.10<br>14  | 63.70<br>14  | 0.01<br>34 | 0.01<br>14   | 0.01<br>14  | —<br>— | 0.02<br>14  |
| gen-w22-h5-rep1.sdvstp  | 0.13<br>16  | 47.80<br>14  | 0.01<br>43 | 0.02<br>14   | 0.01<br>14  | —<br>— | 0.02<br>14  |
| gen-w22-h5-rep2.sdvstp  | 0.24<br>21  | —<br>—       | 0.01<br>45 | 0.01<br>20   | 0.01<br>20  | —<br>— | 0.01<br>20  |
| gen-w22-h5-rep3.sdvstp  | 0.11<br>15  | 161.77<br>15 | 0.01<br>42 | 0.01<br>15   | 0.01<br>15  | —<br>— | 0.02<br>15  |
| gen-w22-h5-rep4.sdvstp  | 0.71<br>24  | —<br>—       | 0.01<br>74 | 0.12<br>23   | 0.01<br>23  | —<br>— | 0.14<br>23  |
| gen-w22-h5-rep5.sdvstp  | 0.45<br>20  | —<br>—       | 0.01<br>57 | 0.02<br>20   | 0.01<br>20  | —<br>— | 0.02<br>20  |
| gen-w22-h6-rep1.sdvstp  | 0.24<br>20  | —<br>—       | 0.01<br>61 | 0.02<br>20   | 0.01<br>20  | —<br>— | 0.04<br>20  |
| gen-w22-h6-rep2.sdvstp  | 0.90<br>23  | —<br>—       | —<br>—     | 0.04<br>21   | 0.01<br>21  | —<br>— | 0.14<br>21  |
| gen-w22-h6-rep3.sdvstp  | 0.15<br>16  | 121.95<br>15 | —<br>—     | 0.02<br>15   | 0.01<br>15  | —<br>— | 0.02<br>15  |
| gen-w22-h6-rep4.sdvstp  | 0.45<br>21  | —<br>—       | 0.01<br>60 | 0.02<br>20   | 0.01<br>20  | —<br>— | 0.04<br>20  |
| gen-w22-h6-rep5.sdvstp  | 0.47<br>21  | —<br>—       | 0.01<br>74 | 0.07<br>20   | 0.02<br>20  | —<br>— | 0.20<br>20  |
| gen-w22-h7-rep1.sdvstp  | 1.40<br>29  | —<br>—       | 0.01<br>80 | 0.12<br>27   | 0.01<br>27  | —<br>— | 1.23<br>27  |
| gen-w22-h7-rep2.sdvstp  | 0.47<br>21  | —<br>—       | 0.01<br>56 | 0.02<br>20   | 0.03<br>20  | —<br>— | 0.03<br>20  |
| gen-w22-h7-rep3.sdvstp  | 0.71<br>24  | —<br>—       | 0.01<br>63 | 0.02<br>23   | 0.01<br>23  | —<br>— | 0.10<br>23  |
| gen-w22-h7-rep4.sdvstp  | 0.19<br>15  | 58.60<br>14  | 0.01<br>65 | 0.02<br>14   | 0.02<br>14  | —<br>— | 0.03<br>14  |
| gen-w22-h7-rep5.sdvstp  | 0.88<br>26  | —<br>—       | 0.01<br>72 | 0.17<br>25   | 0.01<br>25  | —<br>— | 0.56<br>25  |
| gen-w22-h9-rep1.sdvstp  | 2.32<br>33  | —<br>—       | 0.02<br>32 | 112.48<br>32 | 0.01<br>32  | —<br>— | 90.00<br>32 |
| gen-w22-h9-rep2.sdvstp  | 1.27<br>27  | —<br>—       | 0.01<br>25 | 0.74<br>25   | 0.09<br>25  | —<br>— | 48.96<br>25 |
| gen-w22-h9-rep3.sdvstp  | 1.68<br>30  | —<br>—       | 0.01<br>30 | 3.65<br>30   | 0.01<br>30  | —<br>— | 2.05<br>30  |
| gen-w22-h9-rep4.sdvstp  | 1.71<br>31  | —<br>—       | —<br>—     | 6.19<br>30   | 0.03<br>30  | —<br>— | 6.40<br>30  |
| gen-w22-h9-rep5.sdvstp  | 1.89<br>24  | —<br>—       | —<br>—     | 0.86<br>22   | 0.05<br>22  | —<br>— | 3.67<br>22  |
| gen-w26-h11-rep1.sdvstp | 12.84<br>45 | —<br>—       | 0.02<br>42 | —<br>—       | 0.02<br>42  | —<br>— | —<br>—      |
| gen-w26-h11-rep2.sdvstp | 5.72<br>41  | —<br>—       | 0.02<br>40 | 1.28<br>38   | 0.02<br>38  | —<br>— | 57.88<br>38 |
| gen-w26-h11-rep3.sdvstp | 3.53<br>33  | —<br>—       | 0.02<br>32 | —<br>—       | —<br>—      | —<br>— | —<br>—      |
| gen-w26-h11-rep4.sdvstp | 6.45<br>41  | —<br>—       | 0.02<br>39 | 153.52<br>37 | —<br>—      | —<br>— | —<br>—      |
| gen-w26-h11-rep5.sdvstp | 11.14<br>52 | —<br>—       | 0.02<br>48 | —<br>—       | 0.02<br>48  | —<br>— | —<br>—      |
| gen-w26-h14-rep1.sdvstp | 29.35<br>58 | —<br>—       | 0.02<br>52 | —<br>—       | 0.02<br>52  | —<br>— | —<br>—      |
| gen-w26-h14-rep2.sdvstp | 28.23<br>46 | —<br>—       | 0.12<br>46 | —<br>—       | 12.41<br>44 | —<br>— | —<br>—      |
| gen-w26-h14-rep3.sdvstp | 12.67<br>46 | —<br>—       | 0.02<br>46 | —<br>—       | —<br>—      | —<br>— | —<br>—      |
| gen-w26-h14-rep4.sdvstp | 37.53<br>50 | —<br>—       | 0.03<br>48 | —<br>—       | —<br>—      | —<br>— | —<br>—      |
| gen-w26-h14-rep5.sdvstp | 18.74<br>48 | —<br>—       | 0.03<br>49 | —<br>—       | 0.03<br>47  | —<br>— | —<br>—      |
| gen-w26-h18-rep1.sdvstp | 34.17<br>62 | —<br>—       | —<br>—     | —<br>—       | —<br>—      | —<br>— | —<br>—      |
| gen-w26-h18-rep2.sdvstp | 58.07<br>59 | —<br>—       | —<br>—     | —<br>—       | —<br>—      | —<br>— | —<br>—      |
| gen-w26-h18-rep3.sdvstp | 78.66<br>61 | —<br>—       | —<br>—     | —<br>—       | 0.05<br>57  | —<br>— | —<br>—      |
| gen-w26-h18-rep4.sdvstp | 87.87       | —            | —          | —            | —           | —      | —           |

continued on next page



Table 2: runtime and k per instance (continued)

| Instance                | jean   | jonas | karl | lennart | lucas  | pascal | vsevolod |
|-------------------------|--------|-------|------|---------|--------|--------|----------|
|                         | 58     | —     | —    | —       | —      | —      | —        |
| gen-w26-h18-rep5.sdvstp | 99.32  | —     | —    | —       | —      | —      | —        |
|                         | 61     | —     | —    | —       | —      | —      | —        |
| gen-w26-h23-rep1.sdvstp | —      | —     | 0.11 | —       | —      | —      | —        |
|                         | —      | —     | 73   | —       | —      | —      | —        |
| gen-w26-h23-rep2.sdvstp | —      | —     | 0.13 | —       | —      | —      | —        |
|                         | —      | —     | 75   | —       | —      | —      | —        |
| gen-w26-h23-rep3.sdvstp | 133.26 | —     | —    | —       | 293.01 | —      | —        |
|                         | 70     | —     | —    | —       | 66     | —      | —        |
| gen-w26-h23-rep4.sdvstp | —      | —     | —    | —       | 16.82  | —      | —        |
|                         | —      | —     | —    | —       | 79     | —      | —        |
| gen-w26-h23-rep5.sdvstp | —      | —     | —    | —       | 117.57 | —      | —        |
|                         | —      | —     | —    | —       | 80     | —      | —        |
| gen-w26-h3-rep1.sdvstp  | 0.07   | 0.65  | 0.01 | 0.01    | 0.01   | —      | 0.01     |
|                         | 10     | 10    | 26   | 10      | 10     | —      | 10       |
| gen-w26-h3-rep2.sdvstp  | 0.27   | —     | 0.01 | 0.01    | 0.01   | —      | 0.02     |
|                         | 17     | —     | 40   | 17      | 17     | —      | 17       |
| gen-w26-h3-rep3.sdvstp  | 0.07   | 1.40  | 0.01 | 0.01    | 0.01   | —      | 0.01     |
|                         | 11     | 11    | 23   | 11      | 11     | —      | 11       |
| gen-w26-h3-rep4.sdvstp  | 0.07   | 0.42  | 0.01 | 0.01    | 0.02   | —      | 0.01     |
|                         | 9      | 9     | 24   | 9       | 9      | —      | 9        |
| gen-w26-h3-rep5.sdvstp  | 0.10   | 1.45  | 0.01 | 0.01    | 0.01   | —      | 0.01     |
|                         | 11     | 11    | 29   | 11      | 11     | —      | 11       |
| gen-w26-h4-rep1.sdvstp  | 0.33   | —     | 0.01 | 0.01    | 0.01   | —      | 0.02     |
|                         | 17     | —     | 40   | 16      | 16     | —      | 16       |
| gen-w26-h4-rep2.sdvstp  | 0.46   | —     | 0.01 | 0.04    | 0.01   | —      | 0.03     |
|                         | 21     | —     | 54   | 20      | 20     | —      | 20       |
| gen-w26-h4-rep3.sdvstp  | 0.16   | 8.80  | 0.01 | 0.01    | 0.01   | —      | 0.02     |
|                         | 15     | 13    | 36   | 13      | 13     | —      | 13       |
| gen-w26-h4-rep4.sdvstp  | 0.20   | —     | 0.01 | 0.02    | 0.01   | —      | 0.02     |
|                         | 17     | —     | 50   | 17      | 17     | —      | 17       |
| gen-w26-h4-rep5.sdvstp  | 0.30   | —     | 0.01 | 0.02    | 0.02   | —      | 0.03     |
|                         | 20     | —     | 51   | 20      | 20     | —      | 20       |
| gen-w26-h5-rep1.sdvstp  | 1.05   | —     | 0.01 | 0.75    | 0.03   | —      | 1.56     |
|                         | 25     | —     | 85   | 24      | 24     | —      | 24       |
| gen-w26-h5-rep2.sdvstp  | 0.72   | —     | 0.01 | 0.08    | 0.02   | —      | 0.38     |
|                         | 24     | —     | 81   | 22      | 22     | —      | 22       |
| gen-w26-h5-rep3.sdvstp  | 0.78   | —     | —    | 0.04    | 0.03   | —      | 0.04     |
|                         | 21     | —     | —    | 21      | 21     | —      | 21       |
| gen-w26-h5-rep4.sdvstp  | 0.50   | —     | —    | 0.02    | 0.02   | —      | 0.02     |
|                         | 21     | —     | —    | 21      | 21     | —      | 21       |
| gen-w26-h5-rep5.sdvstp  | 0.72   | —     | 0.01 | 0.02    | 0.01   | —      | 0.04     |
|                         | 21     | —     | 67   | 21      | 21     | —      | 21       |
| gen-w26-h6-rep1.sdvstp  | 1.49   | —     | 0.01 | 0.02    | 0.02   | —      | 0.04     |
|                         | 27     | —     | 65   | 26      | 26     | —      | 26       |
| gen-w26-h6-rep2.sdvstp  | 1.01   | —     | 0.01 | 0.10    | 0.01   | —      | 0.40     |
|                         | 26     | —     | 88   | 25      | 25     | —      | 25       |
| gen-w26-h6-rep3.sdvstp  | 0.57   | —     | 0.01 | 0.02    | 0.01   | —      | 0.01     |
|                         | 23     | —     | 51   | 23      | 23     | —      | 23       |
| gen-w26-h6-rep4.sdvstp  | 0.88   | —     | 0.01 | 2.28    | 0.01   | —      | 2.02     |
|                         | 25     | —     | 85   | 25      | 25     | —      | 25       |
| gen-w26-h6-rep5.sdvstp  | 0.21   | —     | 0.01 | 0.19    | 0.03   | —      | 0.15     |
|                         | 18     | —     | 83   | 17      | 17     | —      | 17       |
| gen-w26-h7-rep1.sdvstp  | 0.86   | —     | 0.01 | 0.09    | 0.01   | —      | 0.68     |
|                         | 25     | —     | 89   | 23      | 23     | —      | 23       |
| gen-w26-h7-rep2.sdvstp  | 0.81   | —     | 0.01 | 0.04    | 0.02   | —      | 0.19     |
|                         | 23     | —     | 77   | 20      | 20     | —      | 20       |
| gen-w26-h7-rep3.sdvstp  | 1.38   | —     | 0.01 | 1.14    | 0.02   | —      | 2.76     |
|                         | 26     | —     | 96   | 25      | 25     | —      | 25       |
| gen-w26-h7-rep4.sdvstp  | 0.95   | —     | 0.01 | 0.05    | 0.01   | —      | 0.10     |
|                         | 25     | —     | 78   | 24      | 24     | —      | 24       |
| gen-w26-h7-rep5.sdvstp  | 0.35   | —     | 0.01 | 0.21    | 0.06   | —      | 0.89     |
|                         | 20     | —     | 82   | 19      | 19     | —      | 19       |
| gen-w26-h9-rep1.sdvstp  | 2.76   | —     | 0.01 | 5.05    | 0.02   | —      | 21.77    |
|                         | 33     | —     | 31   | 31      | 31     | —      | 31       |
| gen-w26-h9-rep2.sdvstp  | 4.17   | —     | 0.02 | 1.25    | 0.02   | —      | 114.31   |
|                         | 39     | —     | 35   | 35      | 35     | —      | 35       |
| gen-w26-h9-rep3.sdvstp  | 3.86   | —     | 0.01 | 46.77   | 0.02   | —      | —        |
|                         | 39     | —     | 37   | 37      | 37     | —      | —        |
| gen-w26-h9-rep4.sdvstp  | 2.99   | —     | 0.01 | 2.56    | 0.02   | —      | 8.17     |
|                         | 32     | —     | 32   | 32      | 32     | —      | 32       |

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Table 2: runtime and k per instance (continued)

| Instance                | jean         | jonas        | karl        | lennart    | lucas        | pascal    | vsevolod    |
|-------------------------|--------------|--------------|-------------|------------|--------------|-----------|-------------|
| gen-w26-h9-rep5.sdvstp  | 2.15<br>30   | —<br>—       | 0.02<br>30  | 1.90<br>30 | 0.02<br>30   | —<br>—    | 45.19<br>30 |
| gen-w3-h3-rep1.sdvstp   | 0.06<br>3    | 0.01<br>3    | 0.01<br>3   | —<br>—     | 0.01<br>3    | 0.03<br>3 | 0.01<br>3   |
| gen-w3-h3-rep2.sdvstp   | 0.06<br>2    | 0.02<br>2    | 0.01<br>2   | —<br>—     | 0.01<br>2    | 0.02<br>2 | 0.01<br>2   |
| gen-w3-h3-rep3.sdvstp   | 0.06<br>2    | 0.01<br>2    | 0.01<br>3   | —<br>—     | 0.01<br>2    | 0.02<br>2 | 0.01<br>2   |
| gen-w3-h3-rep4.sdvstp   | 0.06<br>1    | 0.01<br>1    | 0.01<br>5   | —<br>—     | 0.01<br>1    | —<br>—    | 0.01<br>1   |
| gen-w3-h3-rep5.sdvstp   | 0.06<br>4    | 0.02<br>4    | 0.01<br>5   | —<br>—     | 0.01<br>4    | 0.02<br>4 | 0.01<br>4   |
| gen-w31-h11-rep1.sdvstp | 14.77<br>41  | —<br>—       | 0.06<br>40  | —<br>—     | 296.61<br>40 | —<br>—    | —<br>—      |
| gen-w31-h11-rep2.sdvstp | 15.04<br>43  | —<br>—       | 0.03<br>40  | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h11-rep3.sdvstp | 21.64<br>48  | —<br>—       | 0.02<br>48  | —<br>—     | 0.09<br>46   | —<br>—    | —<br>—      |
| gen-w31-h11-rep4.sdvstp | 3.29<br>36   | —<br>—       | 0.03<br>35  | —<br>—     | 0.05<br>32   | —<br>—    | —<br>—      |
| gen-w31-h11-rep5.sdvstp | 15.32<br>42  | —<br>—       | 0.03<br>42  | —<br>—     | 0.02<br>41   | —<br>—    | —<br>—      |
| gen-w31-h14-rep1.sdvstp | 48.99<br>64  | —<br>—       | 0.03<br>63  | —<br>—     | 0.02<br>62   | —<br>—    | —<br>—      |
| gen-w31-h14-rep2.sdvstp | 25.44<br>53  | —<br>—       | 0.03<br>51  | —<br>—     | 0.63<br>51   | —<br>—    | —<br>—      |
| gen-w31-h14-rep3.sdvstp | 93.47<br>73  | —<br>—       | 0.02<br>68  | —<br>—     | 0.15<br>66   | —<br>—    | —<br>—      |
| gen-w31-h14-rep4.sdvstp | 50.55<br>60  | —<br>—       | 0.03<br>58  | —<br>—     | 31.86<br>58  | —<br>—    | —<br>—      |
| gen-w31-h14-rep5.sdvstp | 32.32<br>54  | —<br>—       | 0.13<br>50  | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h18-rep1.sdvstp | 242.25<br>86 | —<br>—       | —<br>—      | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h18-rep2.sdvstp | —<br>—       | —<br>—       | 0.04<br>73  | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h18-rep3.sdvstp | —<br>—       | —<br>—       | 0.06<br>69  | —<br>—     | 196.84<br>69 | —<br>—    | —<br>—      |
| gen-w31-h18-rep4.sdvstp | —<br>—       | —<br>—       | 0.18<br>74  | —<br>—     | 111.16<br>74 | —<br>—    | —<br>—      |
| gen-w31-h18-rep5.sdvstp | —<br>—       | —<br>—       | —<br>—      | —<br>—     | 3.49<br>76   | —<br>—    | —<br>—      |
| gen-w31-h23-rep1.sdvstp | —<br>—       | —<br>—       | —<br>—      | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h23-rep2.sdvstp | —<br>—       | —<br>—       | —<br>—      | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h23-rep3.sdvstp | —<br>—       | —<br>—       | 0.17<br>88  | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h23-rep4.sdvstp | —<br>—       | —<br>—       | 0.83<br>91  | —<br>—     | 74.59<br>90  | —<br>—    | —<br>—      |
| gen-w31-h23-rep5.sdvstp | —<br>—       | —<br>—       | 0.29<br>92  | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h29-rep1.sdvstp | —<br>—       | —<br>—       | —<br>—      | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h29-rep2.sdvstp | —<br>—       | —<br>—       | 0.25<br>110 | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h29-rep3.sdvstp | —<br>—       | —<br>—       | —<br>—      | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h29-rep4.sdvstp | —<br>—       | —<br>—       | —<br>—      | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h29-rep5.sdvstp | —<br>—       | —<br>—       | —<br>—      | —<br>—     | —<br>—       | —<br>—    | —<br>—      |
| gen-w31-h3-rep1.sdvstp  | 0.16<br>15   | 299.74<br>15 | 0.01<br>43  | 0.02<br>15 | 0.01<br>15   | —<br>—    | 0.01<br>15  |
| gen-w31-h3-rep2.sdvstp  | 0.21<br>20   | —<br>—       | 0.01<br>30  | 0.01<br>20 | 0.01<br>20   | —<br>—    | 0.01<br>20  |
| gen-w31-h3-rep3.sdvstp  | 0.13<br>16   | —<br>—       | 0.01<br>52  | 0.02<br>16 | 0.01<br>16   | —<br>—    | 0.02<br>16  |
| gen-w31-h3-rep4.sdvstp  | 0.15<br>16   | —<br>—       | 0.01<br>40  | 0.02<br>16 | 0.01<br>16   | —<br>—    | 0.02<br>16  |
| gen-w31-h3-rep5.sdvstp  | 0.11         | 19.03        | 0.01        | 0.01       | 0.01         | —         | 0.02        |

continued on next page

Table 2: runtime and k per instance (continued)

| Instance               | jean  | jonas | karl | lennart | lucas | pascal | vsevolod |
|------------------------|-------|-------|------|---------|-------|--------|----------|
|                        | 14    | 13    | 41   | 13      | 13    | —      | 13       |
| gen-w31-h4-rep1.sdvstp | 0.57  | —     | 0.01 | 0.02    | 0.01  | —      | 0.05     |
|                        | 21    | —     | 59   | 21      | 21    | —      | 21       |
| gen-w31-h4-rep2.sdvstp | 0.37  | —     | 0.01 | 0.04    | 0.01  | —      | 0.03     |
|                        | 21    | —     | 61   | 20      | 20    | —      | 20       |
| gen-w31-h4-rep3.sdvstp | 0.88  | —     | 0.01 | 0.20    | 0.02  | —      | 0.23     |
|                        | 25    | —     | 79   | 24      | 24    | —      | 24       |
| gen-w31-h4-rep4.sdvstp | 0.25  | —     | 0.01 | 0.03    | 0.01  | —      | 0.05     |
|                        | 20    | —     | 58   | 17      | 17    | —      | 17       |
| gen-w31-h4-rep5.sdvstp | 0.31  | —     | 0.01 | 0.02    | 0.01  | —      | 0.05     |
|                        | 17    | —     | 64   | 17      | 17    | —      | 17       |
| gen-w31-h5-rep1.sdvstp | 0.39  | —     | 0.01 | 0.03    | 0.03  | —      | 0.11     |
|                        | 22    | —     | 65   | 22      | 22    | —      | 22       |
| gen-w31-h5-rep2.sdvstp | 0.85  | —     | —    | 0.55    | 0.02  | —      | 0.74     |
|                        | 22    | —     | —    | 21      | 21    | —      | 21       |
| gen-w31-h5-rep3.sdvstp | 1.43  | —     | 0.01 | 0.40    | 0.01  | —      | 3.17     |
|                        | 26    | —     | 76   | 23      | 23    | —      | 23       |
| gen-w31-h5-rep4.sdvstp | 1.23  | —     | 0.01 | 0.19    | 0.87  | —      | 0.29     |
|                        | 28    | —     | 81   | 25      | 25    | —      | 25       |
| gen-w31-h5-rep5.sdvstp | 0.75  | —     | 0.01 | 0.27    | 0.01  | —      | 0.32     |
|                        | 24    | —     | 90   | 23      | 23    | —      | 23       |
| gen-w31-h6-rep1.sdvstp | 1.10  | —     | 0.01 | 0.81    | 0.07  | —      | 8.17     |
|                        | 26    | —     | 100  | 25      | 25    | —      | 25       |
| gen-w31-h6-rep2.sdvstp | 1.39  | —     | 0.02 | 0.47    | 0.03  | —      | 2.01     |
|                        | 29    | —     | 112  | 28      | 28    | —      | 28       |
| gen-w31-h6-rep3.sdvstp | 1.10  | —     | 0.01 | 0.94    | 0.01  | —      | 1.28     |
|                        | 28    | —     | 99   | 27      | 27    | —      | 27       |
| gen-w31-h6-rep4.sdvstp | 1.89  | —     | 0.01 | 13.44   | 0.02  | —      | 26.53    |
|                        | 32    | —     | 110  | 31      | 31    | —      | 31       |
| gen-w31-h6-rep5.sdvstp | 2.88  | —     | 0.01 | 0.96    | 0.01  | —      | 7.29     |
|                        | 33    | —     | 98   | 32      | 32    | —      | 32       |
| gen-w31-h7-rep1.sdvstp | 2.86  | —     | 0.01 | 0.56    | 0.01  | —      | 5.28     |
|                        | 35    | —     | 35   | 35      | 35    | —      | 35       |
| gen-w31-h7-rep2.sdvstp | 2.63  | —     | 0.02 | —       | 0.40  | —      | —        |
|                        | 33    | —     | 32   | —       | 32    | —      | —        |
| gen-w31-h7-rep3.sdvstp | 4.33  | —     | —    | 54.64   | 0.02  | —      | 9.79     |
|                        | 37    | —     | —    | 36      | 36    | —      | 36       |
| gen-w31-h7-rep4.sdvstp | 2.33  | —     | 0.01 | 0.50    | 0.02  | —      | 11.39    |
|                        | 30    | —     | 29   | 29      | 29    | —      | 29       |
| gen-w31-h7-rep5.sdvstp | 6.75  | —     | —    | 46.91   | 0.02  | —      | —        |
|                        | 38    | —     | —    | 34      | 34    | —      | —        |
| gen-w31-h9-rep1.sdvstp | 4.99  | —     | 0.02 | —       | 0.04  | —      | —        |
|                        | 41    | —     | 39   | —       | 39    | —      | —        |
| gen-w31-h9-rep2.sdvstp | 4.97  | —     | 0.02 | —       | 0.07  | —      | —        |
|                        | 36    | —     | 34   | —       | 33    | —      | —        |
| gen-w31-h9-rep3.sdvstp | 13.02 | —     | 0.02 | —       | 0.05  | —      | —        |
|                        | 42    | —     | 40   | —       | 40    | —      | —        |
| gen-w31-h9-rep4.sdvstp | 3.97  | —     | 0.02 | 123.72  | 0.04  | —      | —        |
|                        | 40    | —     | 37   | 37      | 37    | —      | —        |
| gen-w31-h9-rep5.sdvstp | 3.42  | —     | 0.02 | —       | 0.04  | —      | —        |
|                        | 36    | —     | 34   | —       | 34    | —      | —        |
| gen-w4-h3-rep1.sdvstp  | 0.06  | 0.01  | 0.01 | 0.01    | 0.01  | 0.04   | 0.01     |
|                        | 5     | 5     | 5    | 5       | 5     | 5      | 5        |
| gen-w4-h3-rep2.sdvstp  | 0.06  | 0.01  | 0.01 | —       | 0.01  | 0.04   | 0.01     |
|                        | 4     | 4     | 4    | —       | 4     | 4      | 4        |
| gen-w4-h3-rep3.sdvstp  | 0.06  | 0.01  | 0.01 | 0.01    | 0.01  | 0.04   | 0.01     |
|                        | 4     | 4     | 4    | 4       | 4     | 4      | 4        |
| gen-w4-h3-rep4.sdvstp  | 0.06  | 0.01  | 0.01 | 0.01    | 0.01  | 0.04   | 0.01     |
|                        | 4     | 4     | 4    | 4       | 4     | 4      | 4        |
| gen-w4-h3-rep5.sdvstp  | 0.06  | 0.01  | 0.01 | —       | 0.01  | 0.04   | 0.01     |
|                        | 3     | 3     | 3    | —       | 3     | 3      | 3        |
| gen-w4-h4-rep1.sdvstp  | 0.06  | 0.01  | 0.01 | 0.01    | 0.01  | 0.06   | 0.01     |
|                        | 2     | 2     | 2    | 2       | 2     | 2      | 2        |
| gen-w4-h4-rep2.sdvstp  | 0.06  | 0.01  | 0.01 | 0.01    | 0.01  | 0.06   | 0.01     |
|                        | 4     | 4     | 4    | 4       | 4     | 4      | 4        |
| gen-w4-h4-rep3.sdvstp  | 0.06  | 0.01  | 0.01 | 0.01    | 0.01  | 0.07   | 0.01     |
|                        | 3     | 3     | 7    | 3       | 3     | 3      | 3        |
| gen-w4-h4-rep4.sdvstp  | 0.06  | 0.01  | 0.01 | —       | 0.01  | 0.06   | 0.01     |
|                        | 4     | 4     | 8    | —       | 4     | 4      | 4        |
| gen-w4-h4-rep5.sdvstp  | 0.06  | 0.01  | 0.01 | 0.01    | 0.01  | 0.07   | 0.01     |
|                        | 4     | 4     | 4    | 4       | 4     | 4      | 4        |

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Table 2: runtime and k per instance (continued)

| Instance              | jean      | jonas     | karl       | lennart   | lucas     | pascal     | vsevolod  |
|-----------------------|-----------|-----------|------------|-----------|-----------|------------|-----------|
| gen-w5-h3-rep1.sdvstp | 0.06<br>3 | 0.01<br>3 | 0.01<br>6  | 0.01<br>3 | 0.01<br>3 | 0.05<br>3  | 0.01<br>3 |
| gen-w5-h3-rep2.sdvstp | 0.06<br>4 | 0.01<br>4 | 0.01<br>4  | 0.01<br>4 | 0.01<br>4 | 0.06<br>4  | 0.01<br>4 |
| gen-w5-h3-rep3.sdvstp | 0.06<br>6 | 0.01<br>6 | 0.01<br>7  | 0.01<br>6 | 0.01<br>6 | 0.07<br>6  | 0.01<br>6 |
| gen-w5-h3-rep4.sdvstp | 0.06<br>2 | 0.01<br>2 | 0.01<br>2  | 0.01<br>2 | 0.01<br>2 | 0.05<br>2  | 0.01<br>2 |
| gen-w5-h3-rep5.sdvstp | 0.05<br>3 | 0.01<br>3 | 0.01<br>3  | —<br>—    | 0.01<br>3 | 0.05<br>3  | 0.01<br>3 |
| gen-w5-h4-rep1.sdvstp | 0.06<br>4 | 0.01<br>4 | 0.01<br>4  | 0.01<br>4 | 0.01<br>4 | 0.15<br>4  | 0.01<br>4 |
| gen-w5-h4-rep2.sdvstp | 0.06<br>3 | 0.01<br>3 | 0.01<br>9  | 0.01<br>3 | 0.01<br>3 | 0.16<br>3  | 0.01<br>3 |
| gen-w5-h4-rep3.sdvstp | 0.06<br>4 | 0.01<br>4 | 0.01<br>4  | 0.01<br>4 | 0.01<br>4 | 0.15<br>4  | 0.01<br>4 |
| gen-w5-h4-rep4.sdvstp | 0.06<br>4 | 0.01<br>4 | 0.01<br>13 | 0.01<br>4 | 0.01<br>4 | 0.14<br>4  | 0.01<br>4 |
| gen-w5-h4-rep5.sdvstp | 0.06<br>4 | 0.01<br>4 | 0.01<br>4  | 0.01<br>4 | 0.01<br>4 | 0.13<br>4  | 0.01<br>4 |
| gen-w5-h5-rep1.sdvstp | 0.06<br>5 | 0.01<br>4 | 0.01<br>15 | 0.01<br>4 | 0.02<br>4 | 0.33<br>4  | 0.02<br>4 |
| gen-w5-h5-rep2.sdvstp | 0.06<br>6 | 0.02<br>6 | 0.01<br>6  | 0.01<br>6 | 0.01<br>6 | 0.26<br>6  | 0.01<br>6 |
| gen-w5-h5-rep3.sdvstp | 0.06<br>5 | 0.01<br>4 | 0.01<br>10 | 0.01<br>4 | 0.01<br>4 | 0.25<br>4  | 0.01<br>4 |
| gen-w5-h5-rep4.sdvstp | 0.05<br>6 | 0.02<br>6 | 0.01<br>9  | 0.01<br>6 | 0.01<br>6 | 0.33<br>6  | 0.01<br>6 |
| gen-w5-h5-rep5.sdvstp | 0.06<br>4 | 0.02<br>3 | 0.01<br>18 | 0.01<br>3 | 0.02<br>3 | 0.22<br>3  | 0.01<br>3 |
| gen-w6-h3-rep1.sdvstp | 0.06<br>5 | 0.01<br>5 | 0.01<br>9  | 0.01<br>5 | 0.01<br>5 | 0.15<br>5  | 0.01<br>5 |
| gen-w6-h3-rep2.sdvstp | 0.06<br>4 | 0.01<br>4 | 0.01<br>4  | —<br>—    | 0.01<br>4 | 0.08<br>4  | 0.01<br>4 |
| gen-w6-h3-rep3.sdvstp | 0.06<br>6 | 0.01<br>6 | 0.01<br>6  | —<br>—    | 0.01<br>6 | 0.16<br>6  | 0.01<br>6 |
| gen-w6-h3-rep4.sdvstp | 0.07<br>5 | 0.01<br>5 | 0.01<br>5  | 0.01<br>5 | 0.01<br>5 | 0.11<br>5  | 0.01<br>5 |
| gen-w6-h3-rep5.sdvstp | 0.06<br>4 | 0.01<br>4 | 0.01<br>4  | —<br>—    | 0.01<br>4 | 0.08<br>4  | 0.01<br>4 |
| gen-w6-h4-rep1.sdvstp | 0.06<br>8 | 0.04<br>8 | 0.01<br>10 | 0.01<br>8 | 0.01<br>8 | 3.97<br>8  | 0.01<br>8 |
| gen-w6-h4-rep2.sdvstp | 0.06<br>6 | 0.02<br>6 | 0.01<br>9  | 0.01<br>6 | 0.01<br>6 | 0.38<br>6  | 0.01<br>6 |
| gen-w6-h4-rep3.sdvstp | 0.05<br>5 | 0.02<br>5 | 0.01<br>9  | 0.01<br>5 | 0.01<br>5 | 0.35<br>5  | 0.01<br>5 |
| gen-w6-h4-rep4.sdvstp | 0.06<br>5 | 0.01<br>5 | 0.01<br>10 | 0.01<br>5 | 0.01<br>5 | 0.23<br>5  | 0.01<br>5 |
| gen-w6-h4-rep5.sdvstp | 0.09<br>4 | 0.01<br>4 | 0.01<br>12 | 0.01<br>4 | 0.01<br>4 | 0.23<br>4  | 0.01<br>4 |
| gen-w6-h5-rep1.sdvstp | 0.06<br>7 | 0.03<br>7 | 0.01<br>11 | 0.01<br>7 | 0.01<br>7 | 10.72<br>7 | 0.01<br>7 |
| gen-w6-h5-rep2.sdvstp | 0.06<br>6 | 0.02<br>6 | 0.01<br>11 | 0.01<br>6 | 0.01<br>6 | 1.06<br>6  | 0.01<br>6 |
| gen-w6-h5-rep3.sdvstp | 0.06<br>7 | 0.03<br>7 | 0.01<br>7  | 0.01<br>7 | 0.01<br>7 | 3.31<br>7  | 0.01<br>7 |
| gen-w6-h5-rep4.sdvstp | 0.06<br>4 | 0.02<br>4 | 0.01<br>10 | 0.01<br>4 | 0.01<br>4 | 0.50<br>4  | 0.01<br>4 |
| gen-w6-h5-rep5.sdvstp | 0.06<br>5 | 0.05<br>5 | 0.01<br>12 | 0.01<br>5 | 0.01<br>5 | 1.21<br>5  | 0.01<br>5 |
| gen-w6-h6-rep1.sdvstp | 0.06<br>8 | 0.05<br>7 | 0.01<br>12 | 0.01<br>7 | 0.01<br>7 | 24.55<br>7 | 0.01<br>7 |
| gen-w6-h6-rep2.sdvstp | 0.06<br>4 | 0.02<br>4 | 0.01<br>11 | —<br>—    | 0.01<br>4 | 0.64<br>4  | 0.01<br>4 |
| gen-w6-h6-rep3.sdvstp | 0.06<br>6 | 0.02<br>6 | 0.01<br>13 | —<br>—    | 0.01<br>6 | 1.05<br>6  | 0.01<br>6 |
| gen-w6-h6-rep4.sdvstp | 0.06<br>8 | 0.06<br>8 | 0.01<br>13 | 0.01<br>8 | 0.01<br>8 | 32.70<br>8 | 0.01<br>8 |
| gen-w6-h6-rep5.sdvstp | 0.06<br>7 | 0.08<br>7 | 0.01<br>16 | 0.01<br>7 | 0.01<br>7 | 18.07<br>7 | 0.01<br>7 |
| gen-w7-h3-rep1.sdvstp | 0.06      | 0.02      | 0.01       | 0.01      | 0.01      | 0.14       | 0.01      |

*continued on next page*

Table 2: runtime and k per instance (continued)

| Instance              | jean | jonas | karl | lennart | lucas | pascal | vsevolod |
|-----------------------|------|-------|------|---------|-------|--------|----------|
|                       | 5    | 5     | 5    | 5       | 5     | 5      | 5        |
| gen-w7-h3-rep2.sdvstp | 0.05 | 0.01  | 0.01 | 0.01    | 0.01  | 0.20   | 0.01     |
|                       | 5    | 5     | 5    | 5       | 5     | 5      | 5        |
| gen-w7-h3-rep3.sdvstp | 0.06 | 0.01  | 0.01 | 0.01    | 0.01  | 0.15   | 0.01     |
|                       | 4    | 4     | 10   | 4       | 4     | 4      | 4        |
| gen-w7-h3-rep4.sdvstp | 0.05 | 0.01  | 0.01 | —       | 0.01  | 0.12   | 0.01     |
|                       | 3    | 3     | 3    | —       | 3     | 3      | 3        |
| gen-w7-h3-rep5.sdvstp | 0.06 | 0.01  | 0.01 | —       | 0.01  | 0.14   | 0.01     |
|                       | 5    | 5     | 5    | —       | 5     | 5      | 5        |
| gen-w7-h4-rep1.sdvstp | 0.06 | 0.05  | 0.01 | 0.01    | 0.01  | 1.04   | 0.01     |
|                       | 7    | 7     | 7    | 7       | 7     | 7      | 7        |
| gen-w7-h4-rep2.sdvstp | 0.06 | 0.01  | 0.01 | 0.01    | 0.01  | 0.28   | 0.01     |
|                       | 3    | 3     | 3    | 3       | 3     | 3      | 3        |
| gen-w7-h4-rep3.sdvstp | 0.06 | 0.17  | 0.01 | 0.01    | 0.01  | 204.19 | 0.01     |
|                       | 10   | 9     | 15   | 9       | 9     | 9      | 9        |
| gen-w7-h4-rep4.sdvstp | 0.06 | 0.01  | 0.01 | 0.01    | 0.01  | 0.40   | 0.01     |
|                       | 4    | 4     | 7    | 4       | 4     | 4      | 4        |
| gen-w7-h4-rep5.sdvstp | 0.06 | 0.02  | 0.01 | 0.01    | 0.01  | 0.42   | 0.01     |
|                       | 6    | 5     | 13   | 5       | 5     | 5      | 5        |
| gen-w7-h5-rep1.sdvstp | 0.06 | 0.06  | 0.01 | 0.01    | 0.01  | 29.64  | 0.01     |
|                       | 8    | 8     | 8    | 8       | 8     | 8      | 8        |
| gen-w7-h5-rep2.sdvstp | 0.06 | 0.03  | 0.01 | 0.01    | 0.01  | 0.89   | 0.01     |
|                       | 5    | 5     | 14   | 5       | 5     | 5      | 5        |
| gen-w7-h5-rep3.sdvstp | 0.08 | 0.10  | 0.01 | 0.01    | 0.01  | 146.33 | 0.01     |
|                       | 9    | 9     | 11   | 9       | 9     | 9      | 9        |
| gen-w7-h5-rep4.sdvstp | 0.05 | 0.03  | 0.01 | 0.01    | 0.01  | 0.81   | 0.01     |
|                       | 5    | 5     | 10   | 5       | 5     | 5      | 5        |
| gen-w7-h5-rep5.sdvstp | 0.05 | 0.01  | 0.01 | 0.01    | 0.01  | 0.80   | 0.01     |
|                       | 4    | 4     | 10   | 4       | 4     | 4      | 4        |
| gen-w7-h6-rep1.sdvstp | 0.06 | 0.04  | 0.01 | 0.01    | 0.01  | 1.59   | 0.01     |
|                       | 6    | 6     | 11   | 6       | 6     | 6      | 6        |
| gen-w7-h6-rep2.sdvstp | 0.07 | 0.21  | 0.03 | 0.01    | 0.01  | —      | 0.01     |
|                       | 11   | 10    | 21   | 10      | 10    | —      | 10       |
| gen-w7-h6-rep3.sdvstp | 0.07 | 0.01  | 0.01 | 0.01    | 0.01  | 1.02   | 0.01     |
|                       | 3    | 3     | 9    | 3       | 3     | 3      | 3        |
| gen-w7-h6-rep4.sdvstp | 0.07 | 0.09  | 0.01 | 0.01    | 0.01  | 38.93  | 0.01     |
|                       | 8    | 7     | 25   | 7       | 7     | 7      | 7        |
| gen-w7-h6-rep5.sdvstp | 0.08 | 0.03  | 0.01 | 0.01    | 0.01  | 2.76   | 0.01     |
|                       | 6    | 6     | 11   | 6       | 6     | 6      | 6        |
| gen-w7-h7-rep1.sdvstp | 0.07 | 0.14  | 0.01 | 0.01    | 0.02  | —      | 0.01     |
|                       | 9    | 8     | 25   | 8       | 8     | —      | 8        |
| gen-w7-h7-rep2.sdvstp | 0.06 | 0.09  | 0.01 | 0.01    | 0.02  | 6.90   | 0.01     |
|                       | 6    | 6     | 15   | 6       | 6     | 6      | 6        |
| gen-w7-h7-rep3.sdvstp | 0.08 | 1.91  | 0.01 | 0.02    | 0.01  | —      | 0.01     |
|                       | 11   | 11    | 26   | 11      | 11    | —      | 11       |
| gen-w7-h7-rep4.sdvstp | 0.06 | 0.13  | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                       | 9    | 8     | 19   | 8       | 8     | —      | 8        |
| gen-w7-h7-rep5.sdvstp | 0.07 | 0.21  | 0.01 | 0.01    | 0.02  | —      | 0.01     |
|                       | 8    | 8     | 26   | 8       | 8     | —      | 8        |
| gen-w8-h3-rep1.sdvstp | 0.05 | 0.01  | 0.01 | 0.01    | 0.01  | 0.19   | 0.01     |
|                       | 2    | 2     | 2    | 2       | 2     | 2      | 2        |
| gen-w8-h3-rep2.sdvstp | 0.06 | 0.02  | 0.01 | 0.01    | 0.01  | 0.30   | 0.01     |
|                       | 6    | 6     | 9    | 6       | 6     | 6      | 6        |
| gen-w8-h3-rep3.sdvstp | 0.06 | 0.02  | 0.01 | 0.01    | 0.01  | 0.26   | 0.01     |
|                       | 6    | 6     | 6    | 6       | 6     | 6      | 6        |
| gen-w8-h3-rep4.sdvstp | 0.06 | 0.02  | 0.01 | 0.01    | 0.01  | 0.31   | 0.01     |
|                       | 5    | 5     | 12   | 5       | 5     | 5      | 5        |
| gen-w8-h3-rep5.sdvstp | 0.06 | 0.01  | 0.01 | 0.01    | 0.01  | 0.28   | 0.01     |
|                       | 5    | 5     | 9    | 5       | 5     | 5      | 5        |
| gen-w8-h4-rep1.sdvstp | 0.06 | 0.08  | 0.01 | 0.01    | 0.01  | 212.79 | 0.01     |
|                       | 9    | 9     | 16   | 9       | 9     | 9      | 9        |
| gen-w8-h4-rep2.sdvstp | 0.06 | 0.16  | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                       | 10   | 9     | 15   | 9       | 9     | —      | 9        |
| gen-w8-h4-rep3.sdvstp | 0.05 | 0.01  | 0.01 | 0.01    | 0.01  | 0.43   | 0.01     |
|                       | 3    | 3     | 3    | 3       | 3     | 3      | 3        |
| gen-w8-h4-rep4.sdvstp | 0.06 | 0.02  | 0.01 | 0.01    | 0.01  | 0.84   | 0.01     |
|                       | 6    | 6     | 12   | 6       | 6     | 6      | 6        |
| gen-w8-h4-rep5.sdvstp | 0.06 | 0.04  | 0.01 | 0.01    | 0.01  | 7.41   | 0.01     |
|                       | 7    | 7     | 12   | 7       | 7     | 7      | 7        |
| gen-w8-h5-rep1.sdvstp | 0.06 | 0.02  | 0.03 | 0.01    | 0.01  | 1.77   | 0.01     |
|                       | 5    | 5     | 16   | 5       | 5     | 5      | 5        |

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Table 2: runtime and k per instance (continued)

| Instance              | jean       | jonas      | karl       | lennart    | lucas      | pascal      | vsevolod   |
|-----------------------|------------|------------|------------|------------|------------|-------------|------------|
| gen-w8-h5-rep2.sdvstp | 0.06<br>10 | 0.16<br>10 | 0.01<br>10 | 0.01<br>10 | 0.01<br>10 | —<br>—      | 0.01<br>10 |
| gen-w8-h5-rep3.sdvstp | 0.06<br>6  | 0.05<br>6  | 0.01<br>19 | 0.01<br>6  | 0.01<br>6  | 4.38<br>6   | 0.01<br>6  |
| gen-w8-h5-rep4.sdvstp | 0.06<br>8  | 0.09<br>8  | 0.01<br>17 | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.01<br>8  |
| gen-w8-h5-rep5.sdvstp | 0.06<br>6  | 0.04<br>6  | 0.01<br>21 | 0.01<br>6  | 0.01<br>6  | 4.36<br>6   | 0.01<br>6  |
| gen-w8-h6-rep1.sdvstp | 0.09<br>12 | 0.70<br>11 | 0.01<br>20 | 0.01<br>11 | 0.01<br>11 | —<br>—      | 0.01<br>11 |
| gen-w8-h6-rep2.sdvstp | 0.07<br>8  | 0.10<br>8  | 0.01<br>13 | 0.01<br>8  | 0.01<br>8  | 181.86<br>8 | 0.01<br>8  |
| gen-w8-h6-rep3.sdvstp | 0.08<br>6  | 0.06<br>6  | 0.01<br>17 | 0.01<br>6  | 0.01<br>6  | 38.71<br>6  | 0.01<br>6  |
| gen-w8-h6-rep4.sdvstp | 0.07<br>9  | 0.45<br>9  | 0.01<br>25 | 0.01<br>9  | 0.01<br>9  | —<br>—      | 0.01<br>9  |
| gen-w8-h6-rep5.sdvstp | 0.07<br>11 | 0.76<br>11 | 0.01<br>24 | 0.01<br>11 | 0.01<br>11 | —<br>—      | 0.01<br>11 |
| gen-w8-h7-rep1.sdvstp | 0.06<br>8  | 0.07<br>8  | 0.01<br>17 | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.02<br>8  |
| gen-w8-h7-rep2.sdvstp | 0.06<br>8  | 0.16<br>8  | 0.01<br>20 | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.01<br>8  |
| gen-w8-h7-rep3.sdvstp | 0.07<br>11 | 1.64<br>11 | 0.01<br>25 | 0.01<br>11 | 0.01<br>11 | —<br>—      | 0.01<br>11 |
| gen-w8-h7-rep4.sdvstp | 0.09<br>12 | 1.76<br>11 | 0.01<br>38 | 0.02<br>11 | 0.01<br>11 | —<br>—      | 0.02<br>11 |
| gen-w8-h7-rep5.sdvstp | 0.06<br>8  | 0.07<br>8  | 0.01<br>21 | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.01<br>8  |
| gen-w9-h3-rep1.sdvstp | 0.07<br>6  | 0.02<br>6  | 0.01<br>13 | 0.01<br>6  | 0.01<br>6  | 0.99<br>6   | 0.01<br>6  |
| gen-w9-h3-rep2.sdvstp | 0.06<br>8  | 0.03<br>8  | 0.01<br>11 | 0.01<br>8  | 0.01<br>8  | 2.87<br>8   | 0.01<br>8  |
| gen-w9-h3-rep3.sdvstp | 0.06<br>6  | 0.02<br>5  | 0.01<br>13 | 0.01<br>5  | 0.01<br>5  | 0.64<br>5   | 0.01<br>5  |
| gen-w9-h3-rep4.sdvstp | 0.06<br>7  | 0.03<br>7  | 0.01<br>7  | 0.01<br>7  | 0.01<br>7  | 1.63<br>7   | 0.01<br>7  |
| gen-w9-h3-rep5.sdvstp | 0.06<br>4  | 0.02<br>4  | 0.01<br>15 | 0.02<br>4  | 0.01<br>4  | 0.35<br>4   | 0.01<br>4  |
| gen-w9-h4-rep1.sdvstp | 0.06<br>6  | 0.03<br>6  | 0.01<br>14 | 0.01<br>6  | 0.01<br>6  | 2.09<br>6   | 0.01<br>6  |
| gen-w9-h4-rep2.sdvstp | 0.06<br>6  | 0.02<br>6  | 0.01<br>11 | 0.01<br>6  | 0.01<br>6  | 3.61<br>6   | 0.01<br>6  |
| gen-w9-h4-rep3.sdvstp | 0.06<br>8  | 0.03<br>7  | 0.01<br>13 | 0.01<br>7  | 0.01<br>7  | 5.85<br>7   | 0.01<br>7  |
| gen-w9-h4-rep4.sdvstp | 0.06<br>5  | 0.02<br>5  | 0.01<br>8  | 0.01<br>5  | 0.01<br>5  | 0.85<br>5   | 0.01<br>5  |
| gen-w9-h4-rep5.sdvstp | 0.07<br>10 | 0.33<br>10 | 0.01<br>20 | 0.01<br>10 | 0.01<br>10 | —<br>—      | 0.01<br>10 |
| gen-w9-h5-rep1.sdvstp | 0.06<br>9  | 0.22<br>9  | 0.01<br>19 | 0.01<br>9  | 0.01<br>9  | —<br>—      | 0.01<br>9  |
| gen-w9-h5-rep2.sdvstp | 0.06<br>8  | 0.19<br>8  | 0.01<br>16 | 0.01<br>8  | 0.01<br>8  | 86.14<br>8  | 0.01<br>8  |
| gen-w9-h5-rep3.sdvstp | 0.06<br>8  | 0.13<br>8  | 0.01<br>23 | 0.01<br>8  | 0.02<br>8  | —<br>—      | 0.01<br>8  |
| gen-w9-h5-rep4.sdvstp | 0.06<br>5  | 0.02<br>5  | 0.01<br>13 | 0.01<br>5  | 0.01<br>5  | 1.54<br>5   | 0.01<br>5  |
| gen-w9-h5-rep5.sdvstp | 0.06<br>8  | 0.16<br>8  | 0.01<br>16 | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.01<br>8  |
| gen-w9-h6-rep1.sdvstp | 0.08<br>8  | 0.14<br>8  | 0.01<br>16 | 0.01<br>8  | 0.01<br>8  | —<br>—      | 0.01<br>8  |
| gen-w9-h6-rep2.sdvstp | 0.07<br>9  | 0.34<br>9  | 0.01<br>22 | 0.01<br>9  | 0.01<br>9  | —<br>—      | 0.01<br>9  |
| gen-w9-h6-rep3.sdvstp | 0.11<br>14 | —<br>—     | 0.01<br>22 | 0.01<br>14 | 0.01<br>14 | —<br>—      | 0.01<br>14 |
| gen-w9-h6-rep4.sdvstp | 0.07<br>11 | 0.85<br>11 | 0.01<br>23 | 0.01<br>11 | 0.01<br>11 | —<br>—      | 0.01<br>11 |
| gen-w9-h6-rep5.sdvstp | 0.09<br>11 | 1.13<br>11 | 0.01<br>21 | 0.01<br>11 | 0.01<br>11 | —<br>—      | 0.01<br>11 |
| gen-w9-h7-rep1.sdvstp | 0.08<br>12 | 1.08<br>11 | 0.01<br>35 | 0.01<br>11 | 0.01<br>11 | —<br>—      | 0.02<br>11 |
| gen-w9-h7-rep2.sdvstp | 0.07       | 0.95       | 0.01       | 0.01       | 0.01       | —           | 0.01       |

*continued on next page*

Table 2: runtime and k per instance (continued)

| Instance              | jean | jonas | karl | lennart | lucas | pascal | vsevolod |
|-----------------------|------|-------|------|---------|-------|--------|----------|
|                       | 11   | 11    | 22   | 11      | 11    | —      | 11       |
| gen-w9-h7-rep3.sdvstp | 0.09 | 11.62 | 0.01 | 0.02    | 0.01  | —      | 0.02     |
|                       | 13   | 12    | 33   | 12      | 12    | —      | 12       |
| gen-w9-h7-rep4.sdvstp | 0.11 | 8.90  | 0.01 | 0.01    | 0.01  | —      | 0.02     |
|                       | 13   | 13    | 38   | 13      | 13    | —      | 13       |
| gen-w9-h7-rep5.sdvstp | 0.10 | 1.83  | 0.01 | 0.02    | 0.01  | —      | 0.02     |
|                       | 11   | 11    | 38   | 11      | 11    | —      | 11       |
| gen-w9-h9-rep1.sdvstp | 0.07 | 0.72  | 0.01 | 0.02    | 0.01  | —      | 0.01     |
|                       | 11   | 10    | 30   | 10      | 10    | —      | 10       |
| gen-w9-h9-rep2.sdvstp | 0.14 | —     | 0.01 | 0.01    | 0.01  | —      | 0.02     |
|                       | 17   | —     | 38   | 16      | 16    | —      | 16       |
| gen-w9-h9-rep3.sdvstp | 0.09 | 29.44 | 0.01 | 0.01    | 0.01  | —      | 0.01     |
|                       | 13   | 13    | 31   | 13      | 13    | —      | 13       |
| gen-w9-h9-rep4.sdvstp | 0.12 | 44.30 | 0.01 | 0.03    | 0.02  | —      | 0.03     |
|                       | 14   | 13    | 53   | 13      | 13    | —      | 13       |
| gen-w9-h9-rep5.sdvstp | 0.14 | —     | 0.01 | 0.01    | 0.01  | —      | 0.02     |
|                       | 17   | —     | 37   | 17      | 17    | —      | 17       |